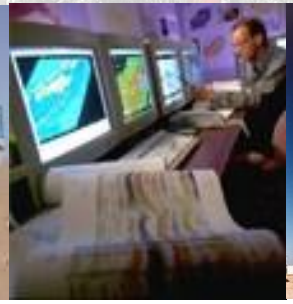
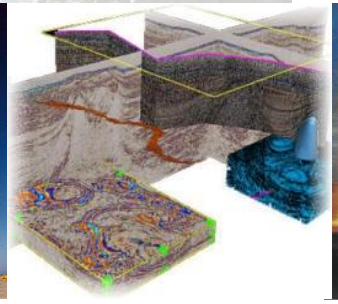
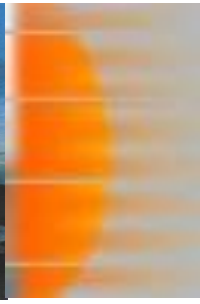
ENG-3020
RENEWABLE ENERGY SOURCES







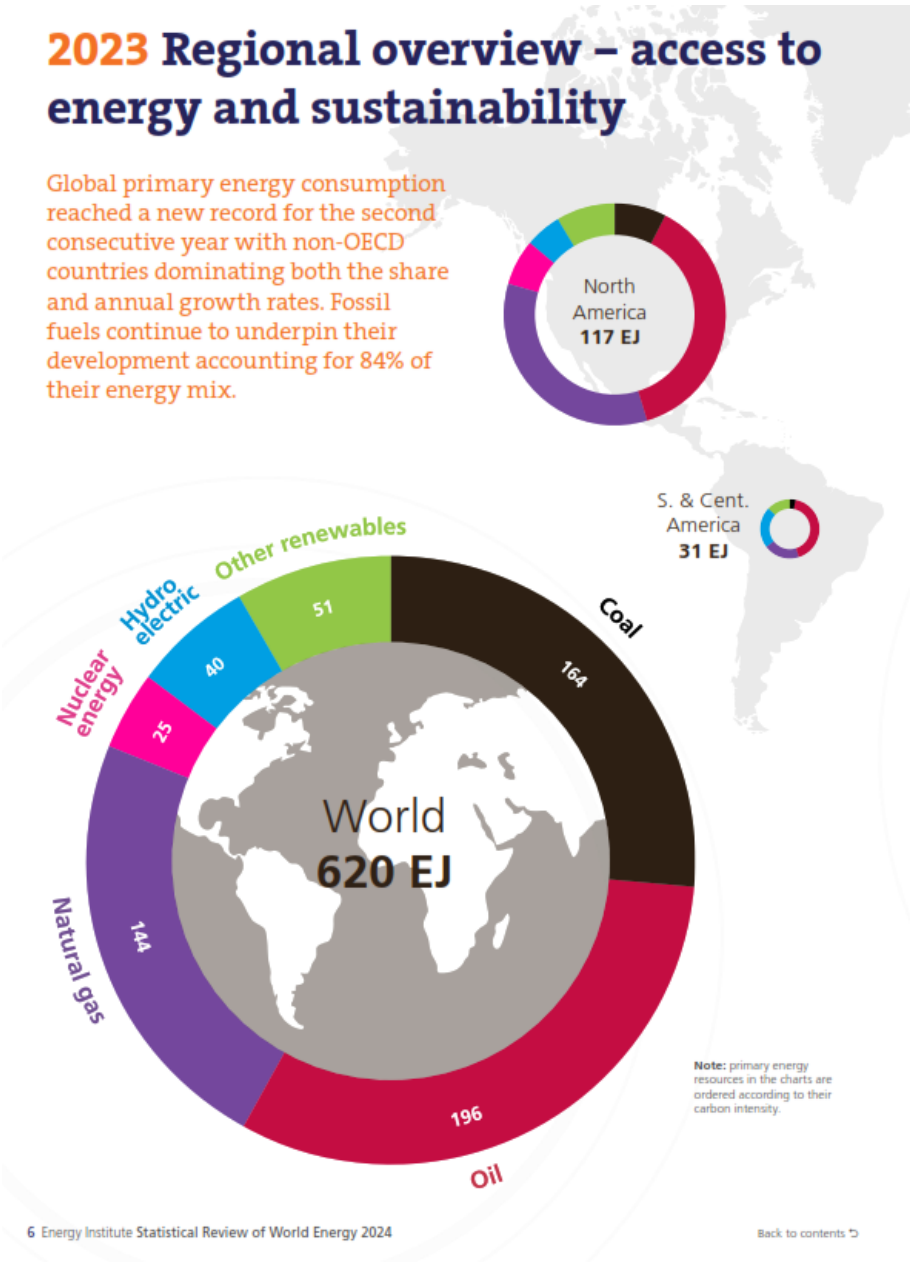


ENERGY BALANCE
BETWEEN
PRODUCTION and CONSUMPTION

Energy Consumption in the World

2023 Regional overview – access to energy and sustainability

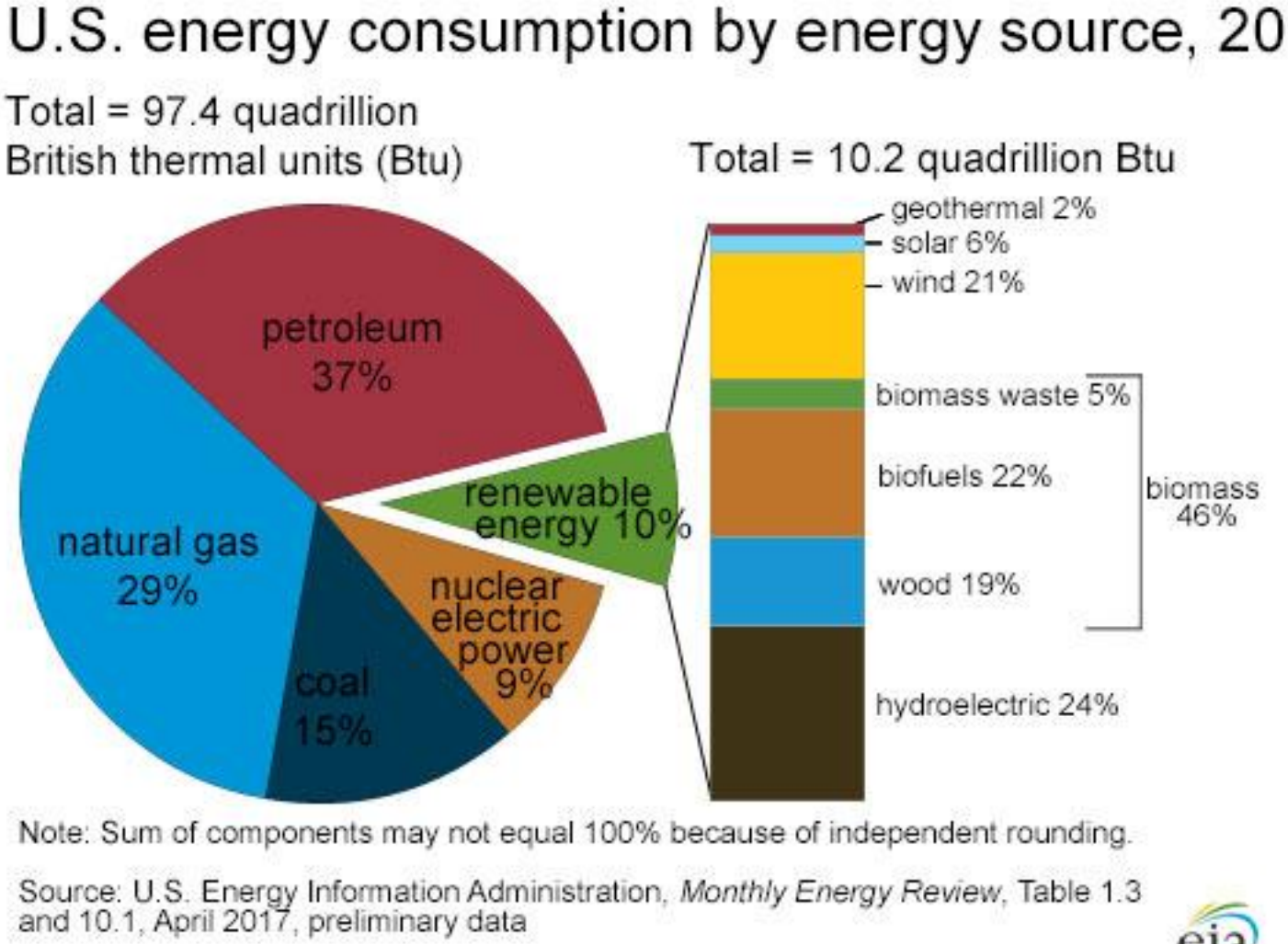
Global primary energy consumption reached a new record for the second consecutive year with non-OECD countries dominating both the share and annual growth rates. Fossil fuels continue to underpin their development accounting for 84% of their energy mix.



Note: primary energy resources in the charts are ordered according to their carbon intensity.

Back to contents

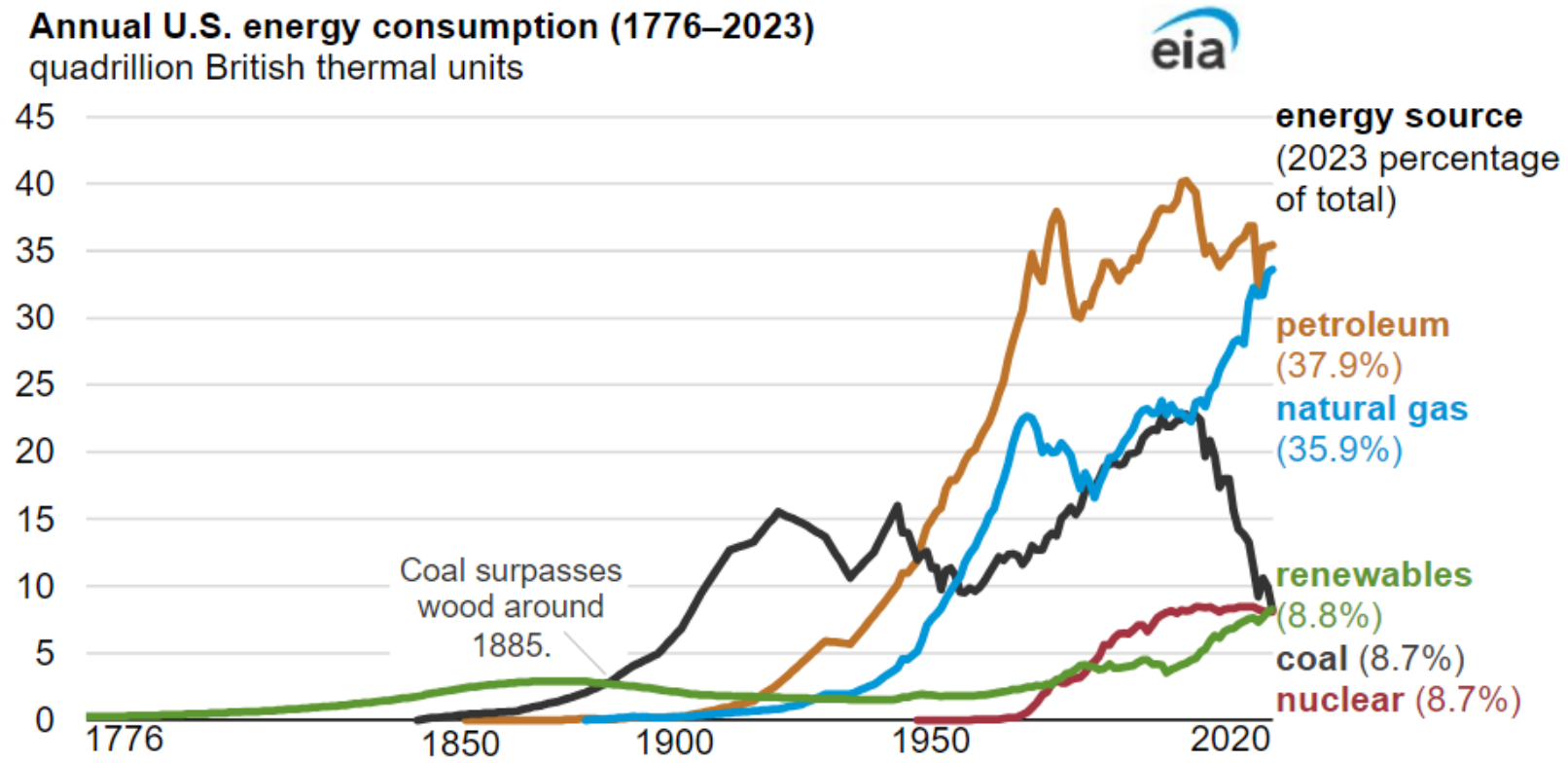
Energy Consumption in USA



Energy Consumption in USA

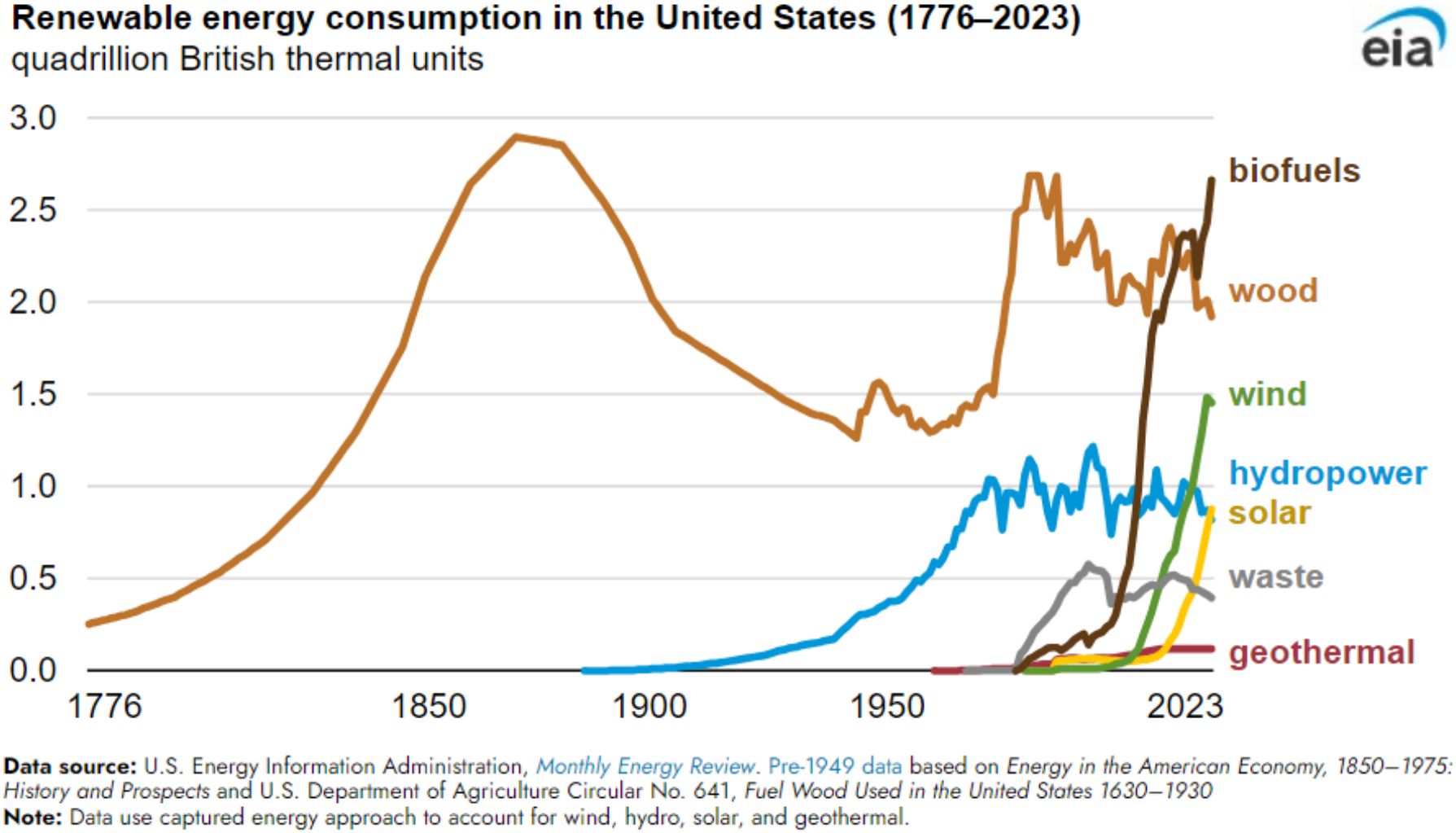
JULY 3, 2024

How has energy use changed throughout U.S. history?

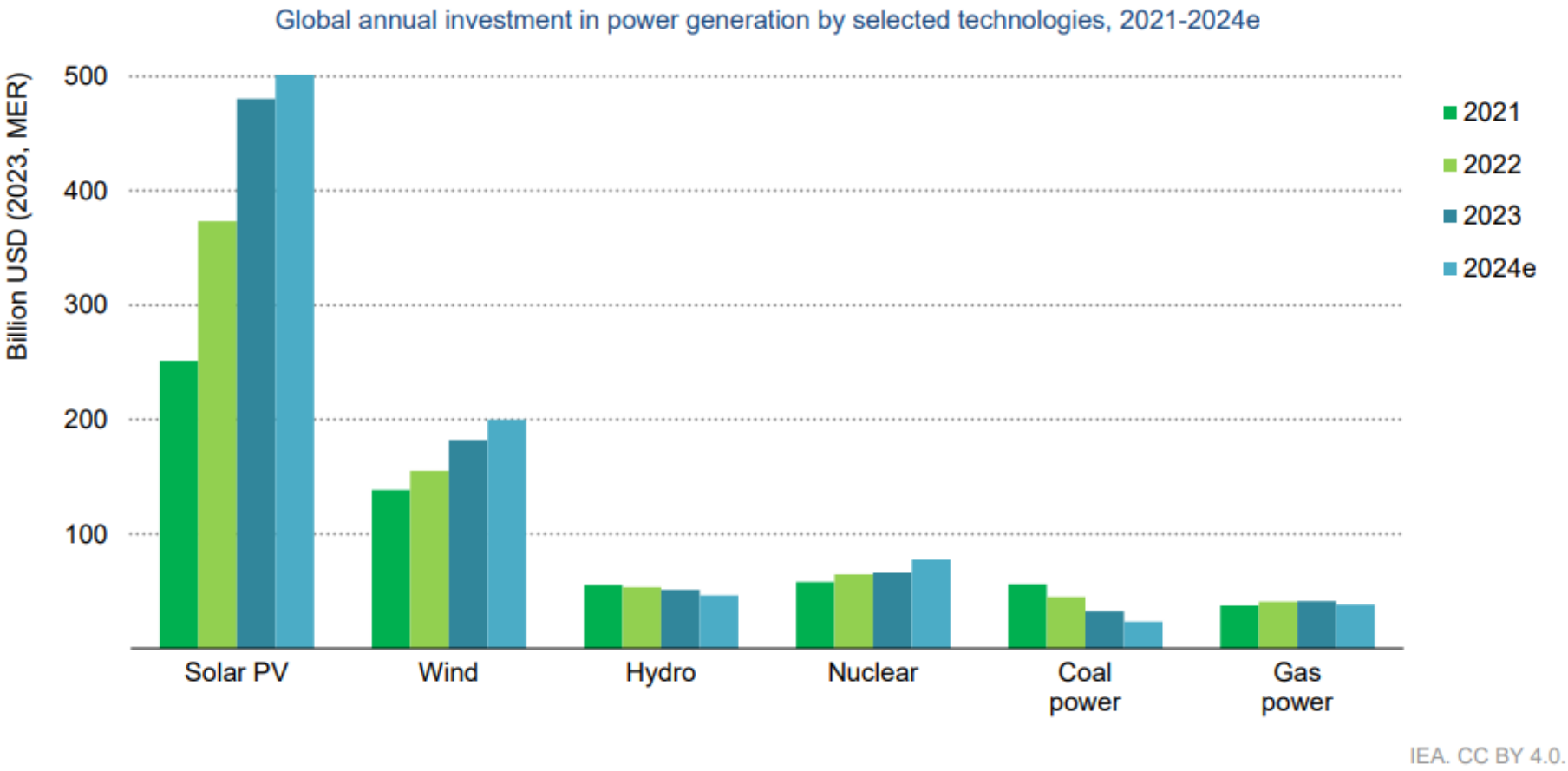


Data source: U.S. Energy Information Administration, *Monthly Energy Review*. Pre-1949 data based on *Energy in the American Economy, 1850–1975: Its History and Prospects* and U.S. Department of Agriculture Circular No. 641, *Fuel Wood Used in the United States 1630–1930*
Note: Data use captured energy approach to account for wind, hydro, solar, and geothermal.

Renewable Energy Consumption in USA



Solar PV attracted a record USD 480 billion in spending in 2023 – more than all other generation technologies combined – while investment in coal power has fallen by 40% since 2021

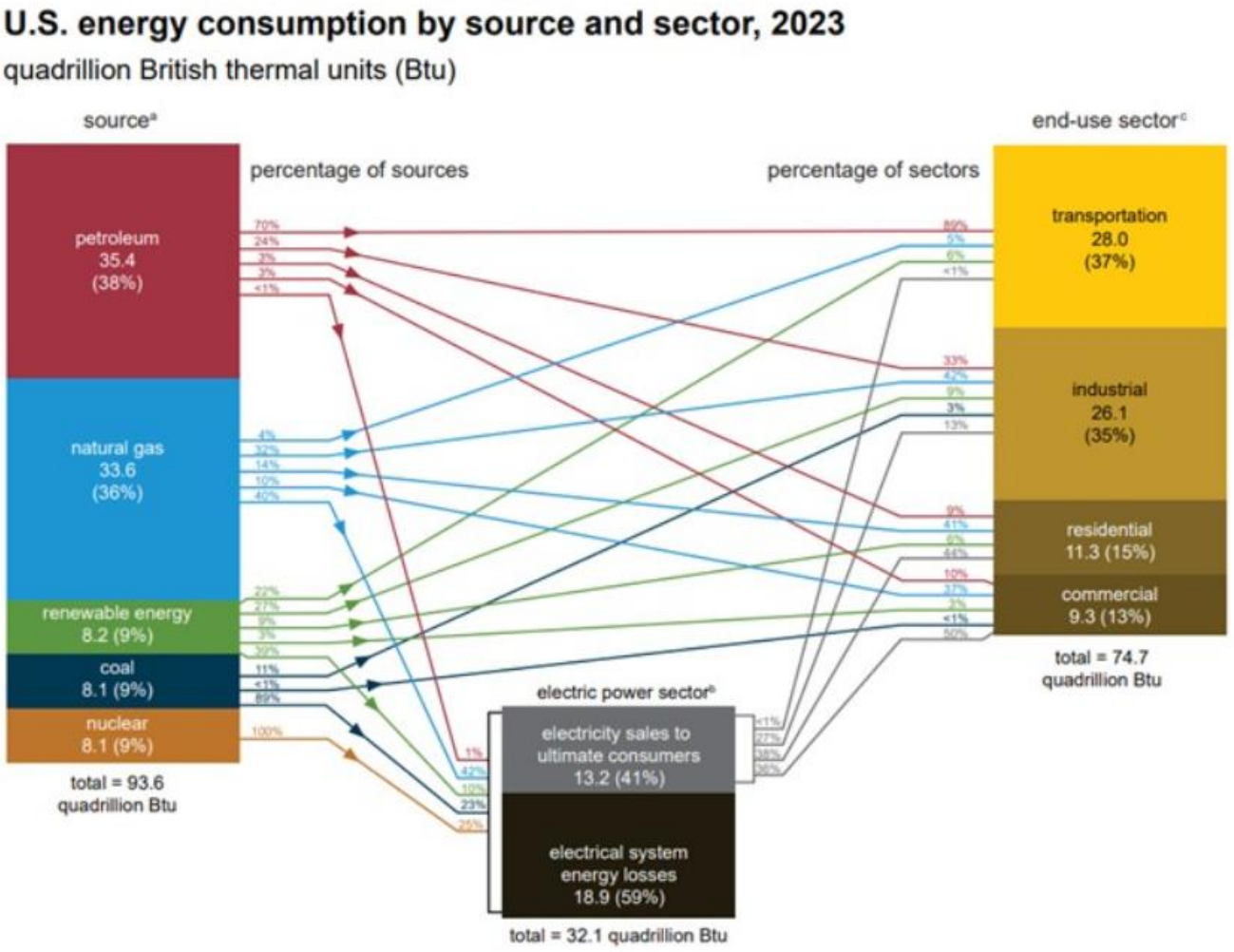


IEA. CC BY 4.0.

Note: Gas-fired generation investment includes both large-scale plants and small-scale generating sets and engines. Hydropower includes pumped-hydro storage. 2024e = estimated values for 2024.

IEA Report-World Energy Investment 2024

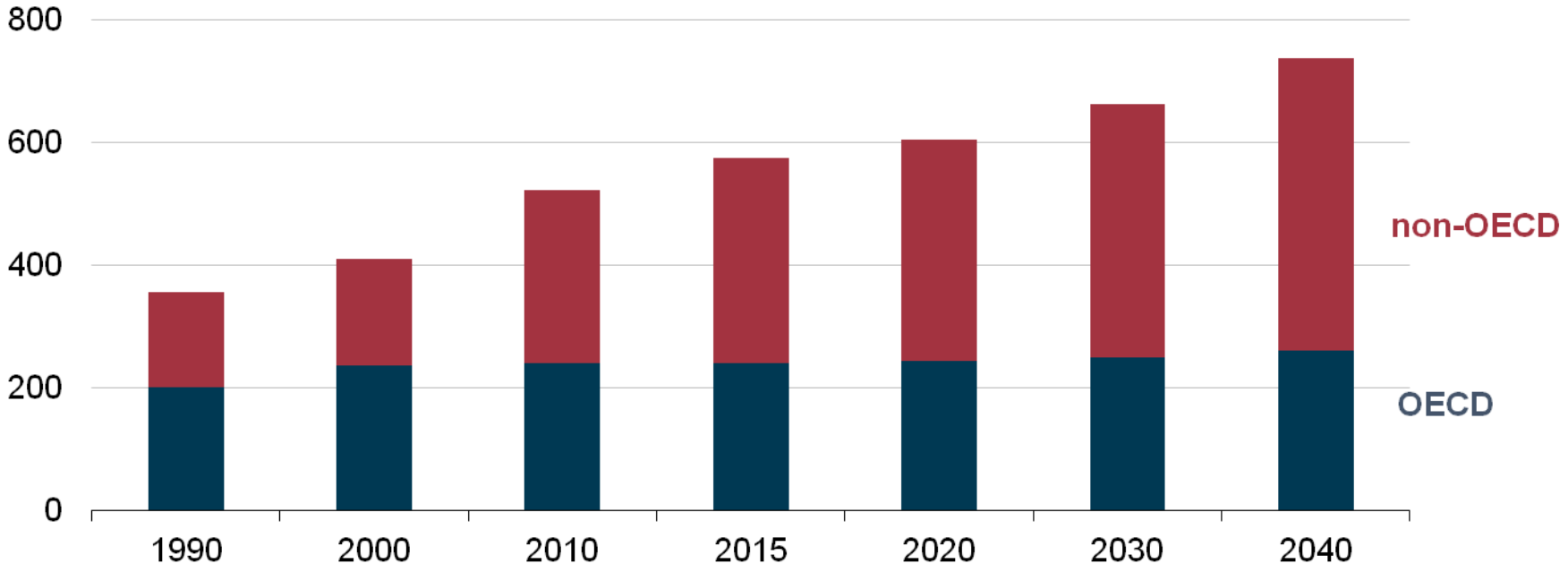
U.S. Energy consumption by source and sector



Data source: U.S. Energy Information Administration, *Monthly Energy Review*
Click to see full chart.

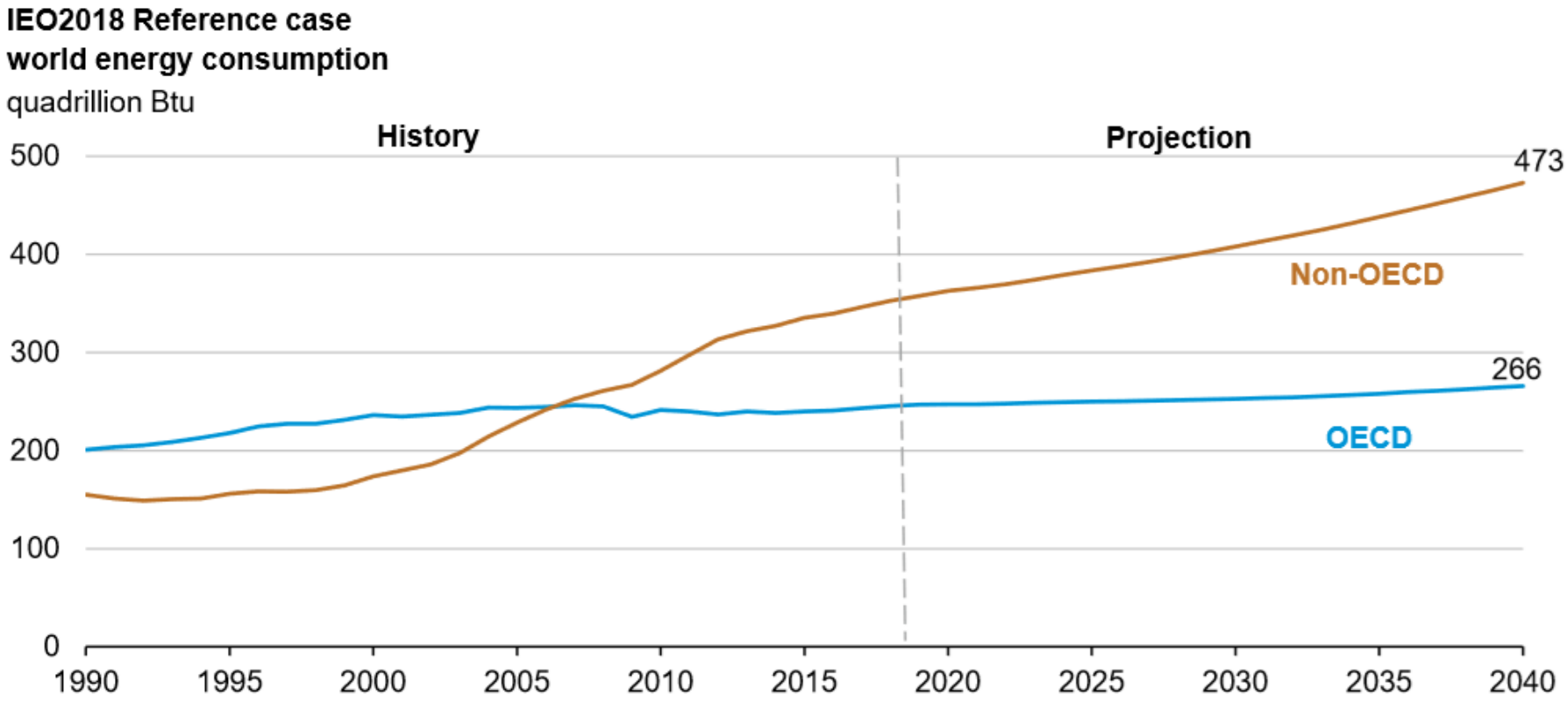
World energy consumption between 2015 and 2040

- **World energy consumption**
- quadrillion Btu



IEO-2017, EIA

World energy consumption between 1990 and 2040



Source: EIA, International Energy Outlook 2018

OECD COUNTRIES

FOUNDING MEMBER STATES

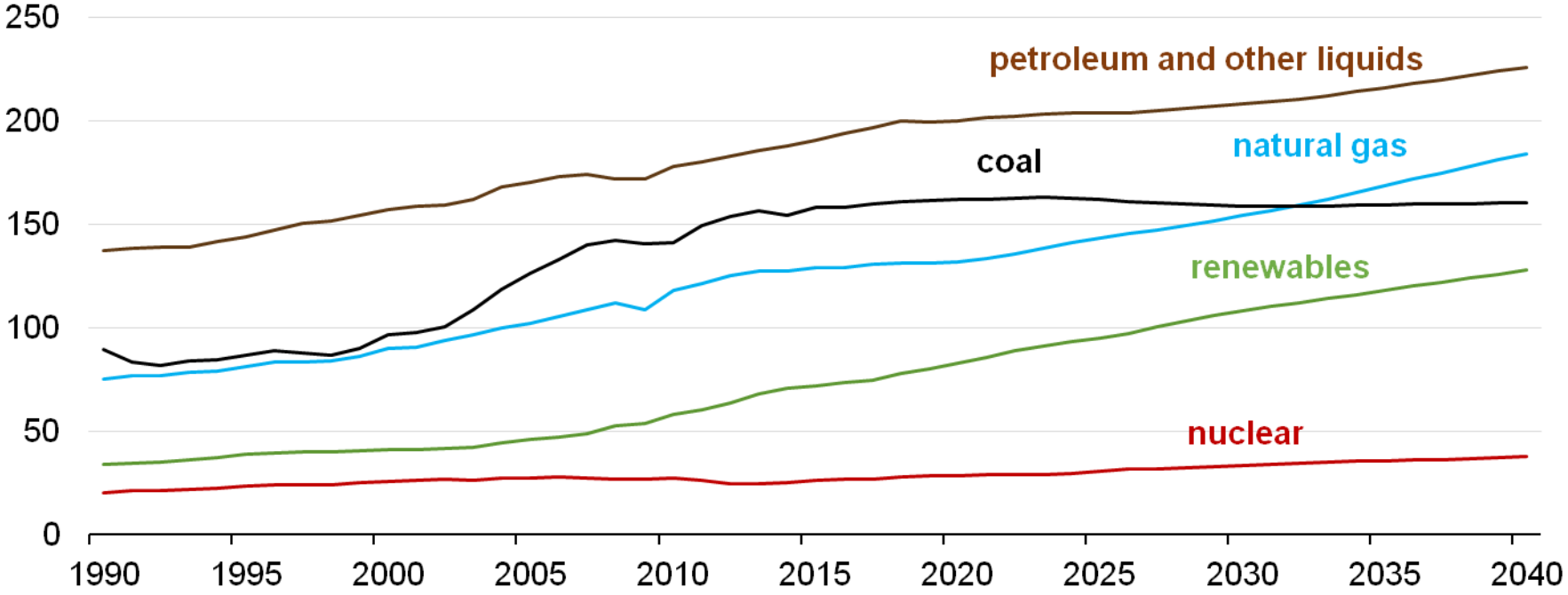
Austria	USA	UK	Belgium	Canada
Denmark	France	Germany	Greece	Iceland
Ireland	Italy	Luxemburg	Netherland	Norway
Portugal	Spain	Sweden	Switzerland	Turkey

POST-OECD COUNTRIES

Australia	Czech Rep.	Findland	Hungary	Japan
Mexico	N. Zealand	Poland	Slovakia	S. Korea
Chile	Estonia	Slovenia	Israel	

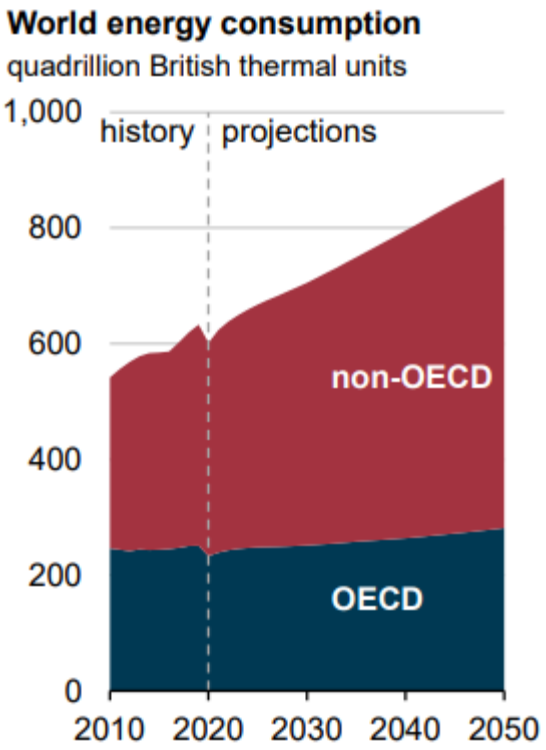
Energy Consumption in the World by source

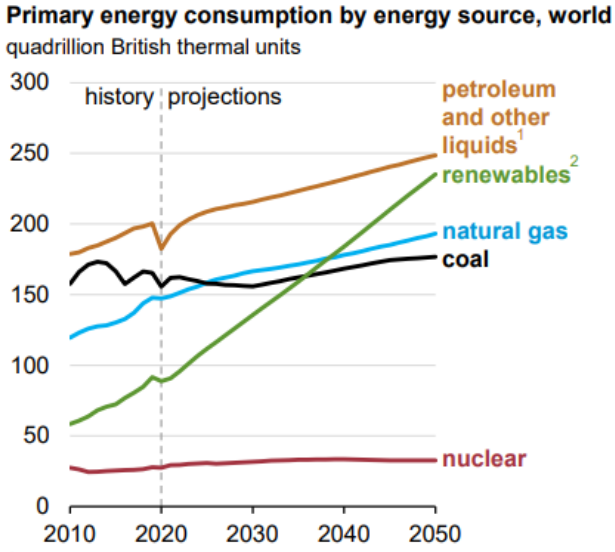
- **World energy consumption by energy source**
- quadrillion Btu

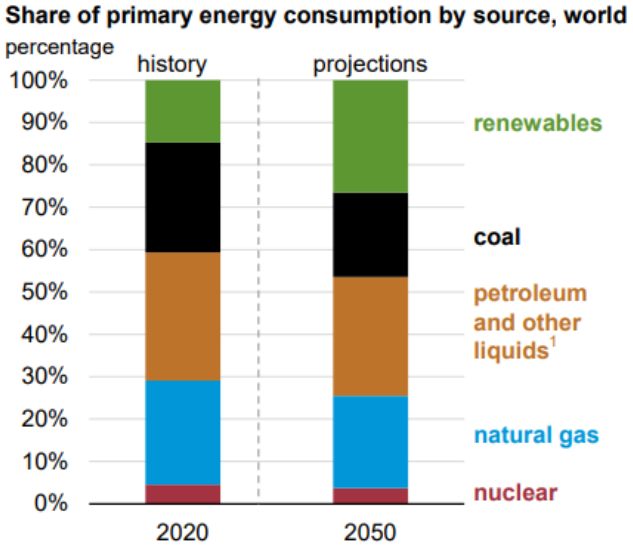


IEO-2017, EIA

IEO-2021 Release Report

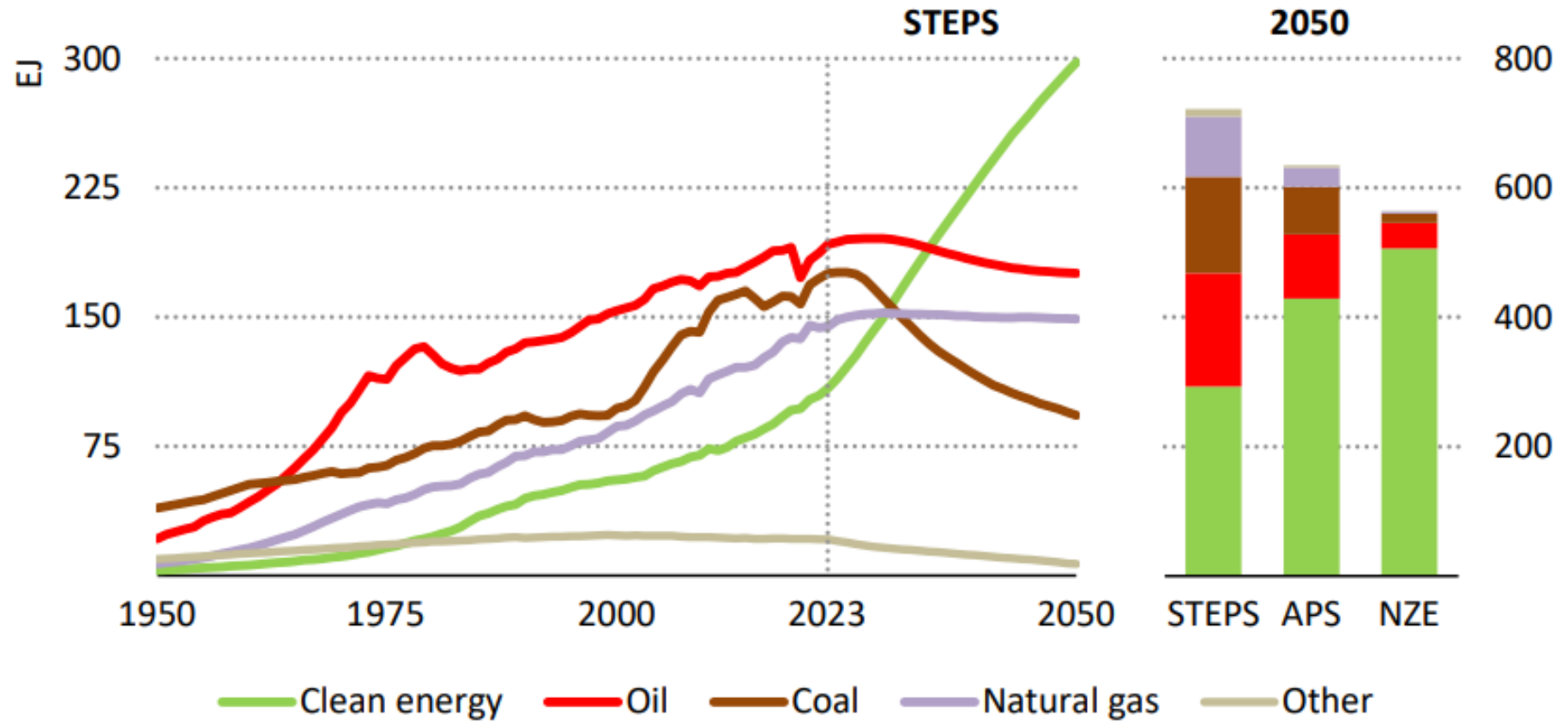




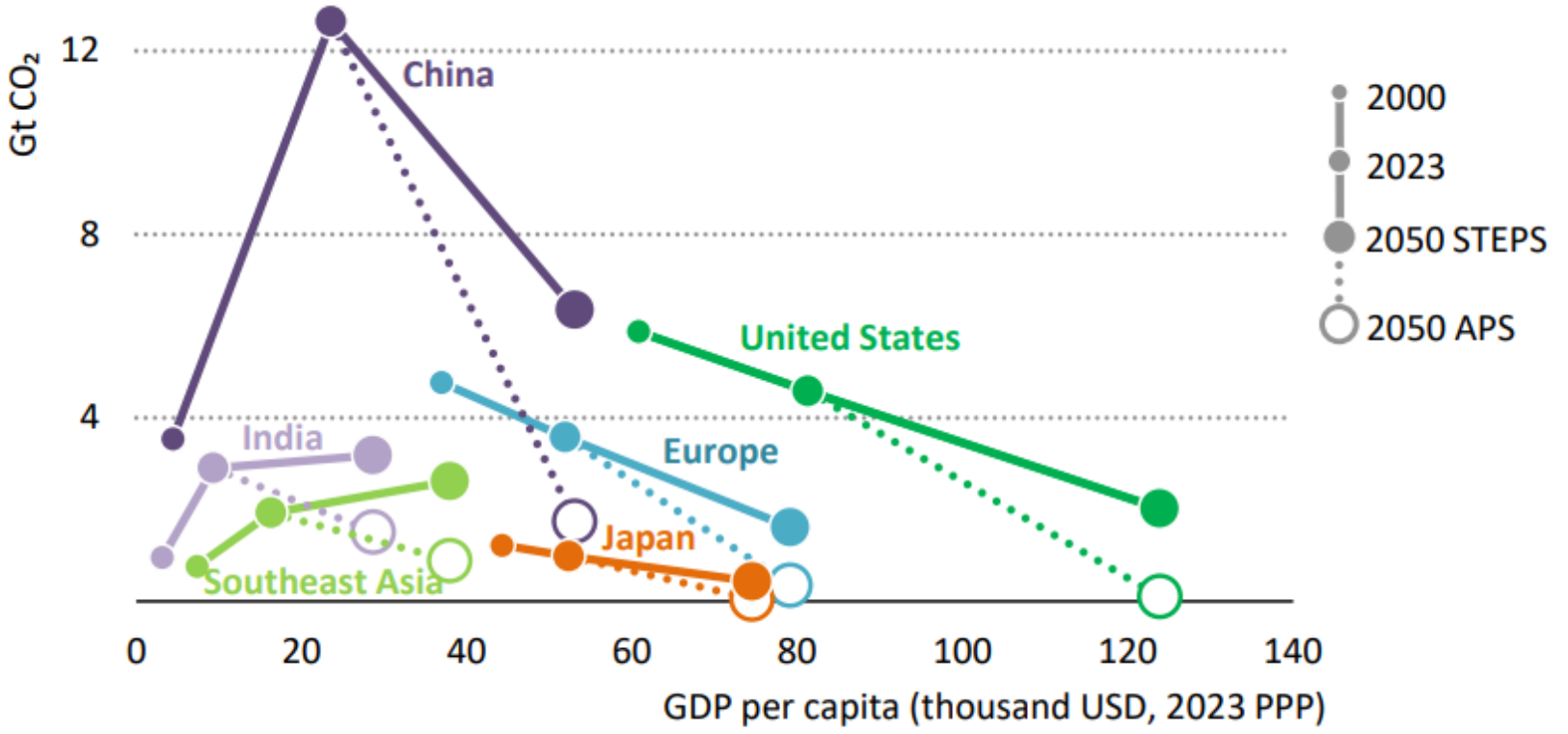


[1] *Includes biofuels*
[2] *Electricity generation from renewable sources is converted to Btu at a rate of 8,124 Btu/kWh*

World Energy Outlook 2024



CO2 emissions in selected countries/regions in the STEPS and APS



IEA. CC BY 4.0

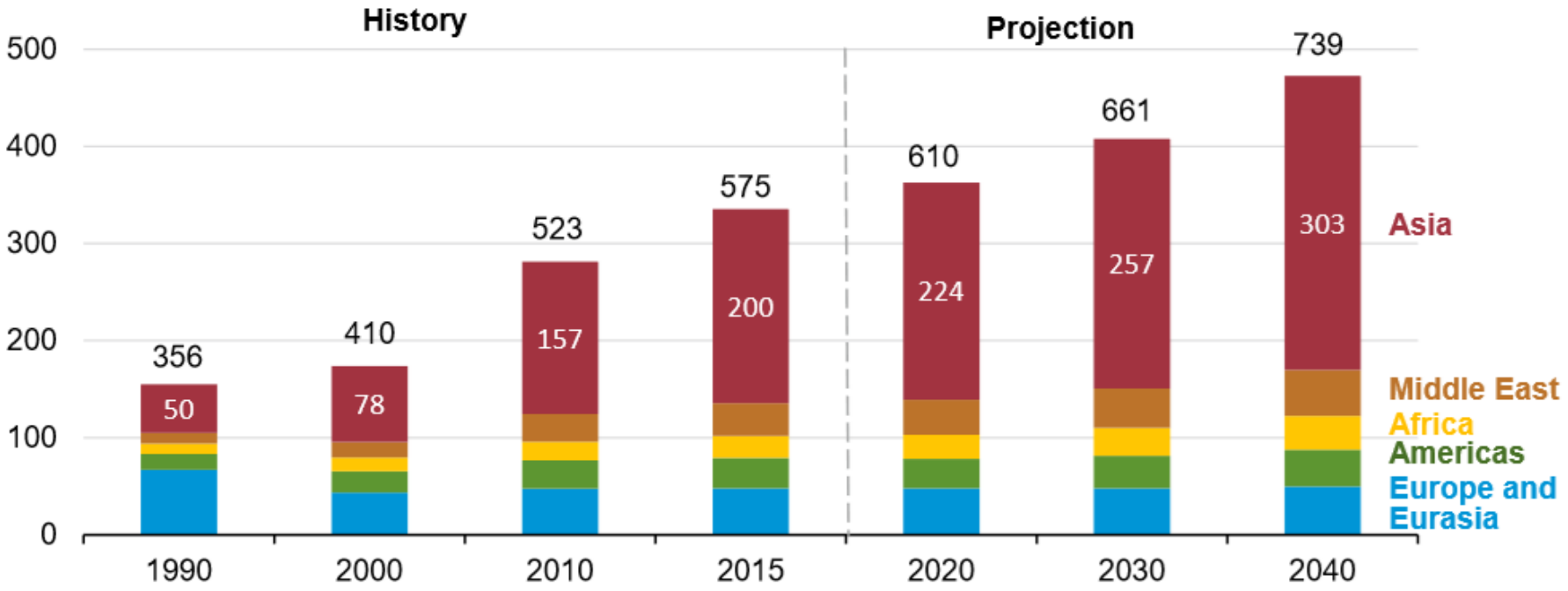
Global CO_2 emissions peak and gradually decline to 2050, led by sharp drops in China as well as in advanced economies

Note: Gt CO_2= gigatonnes of carbon dioxide.

CC BY 4.0.

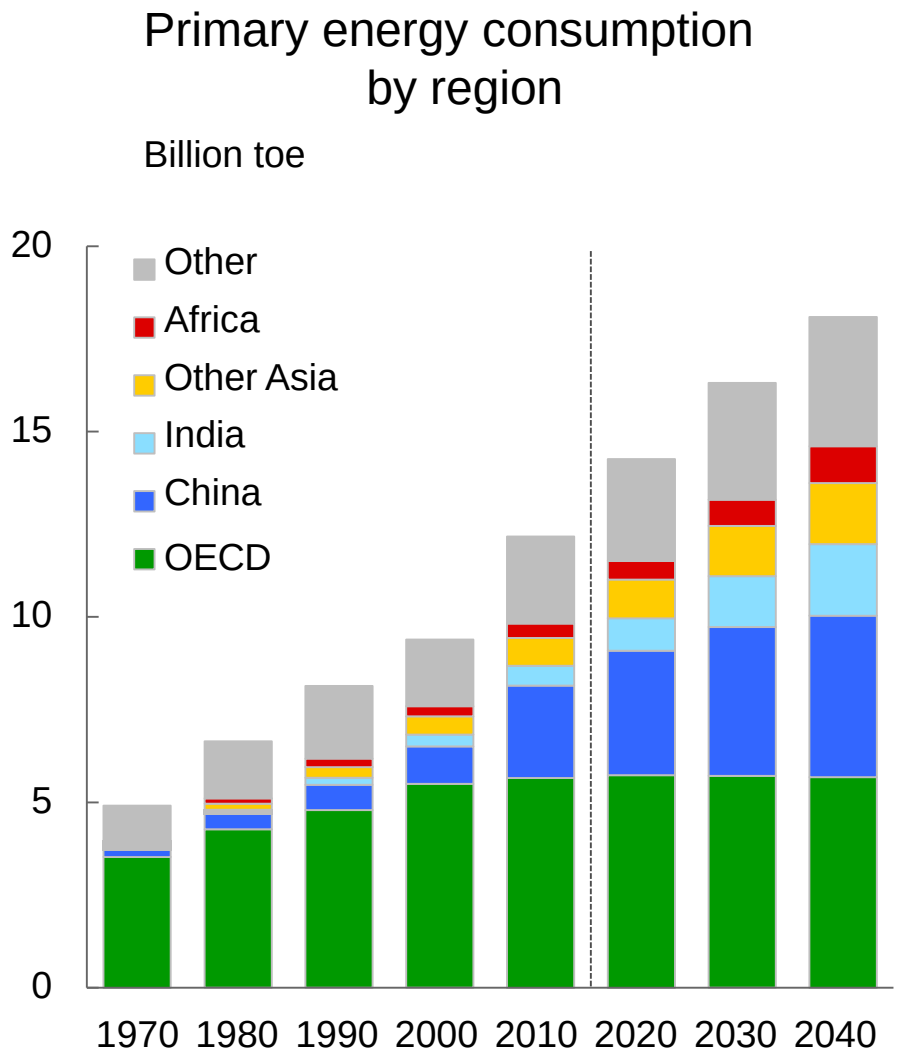
World energy consumption by region between 1990 and 2040

IEO2018 Reference case
non-OECD energy consumption by region
quadrillion Btu



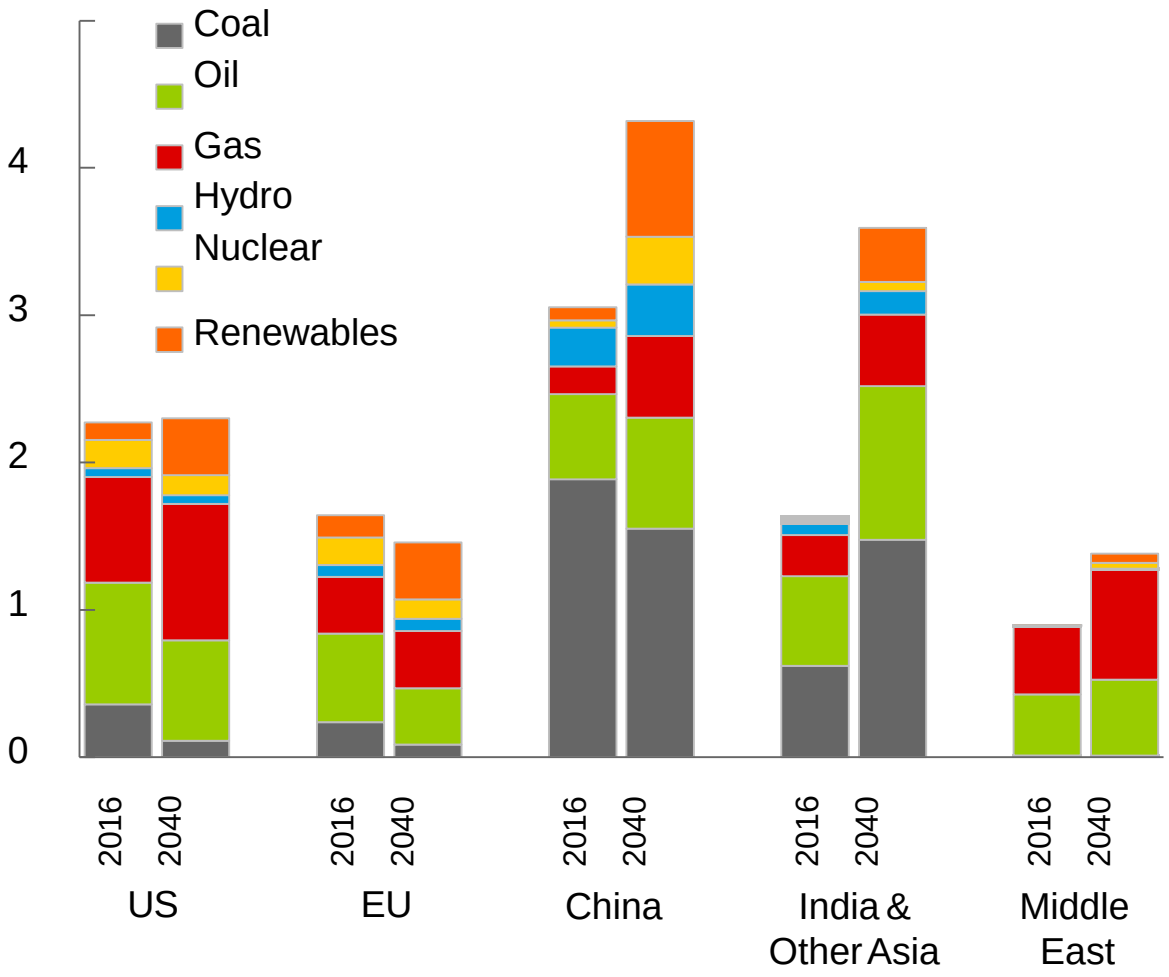
Source: EIA, International Energy Outlook 2018

Energy Consumption in the world



BP Energy Outlook-2018

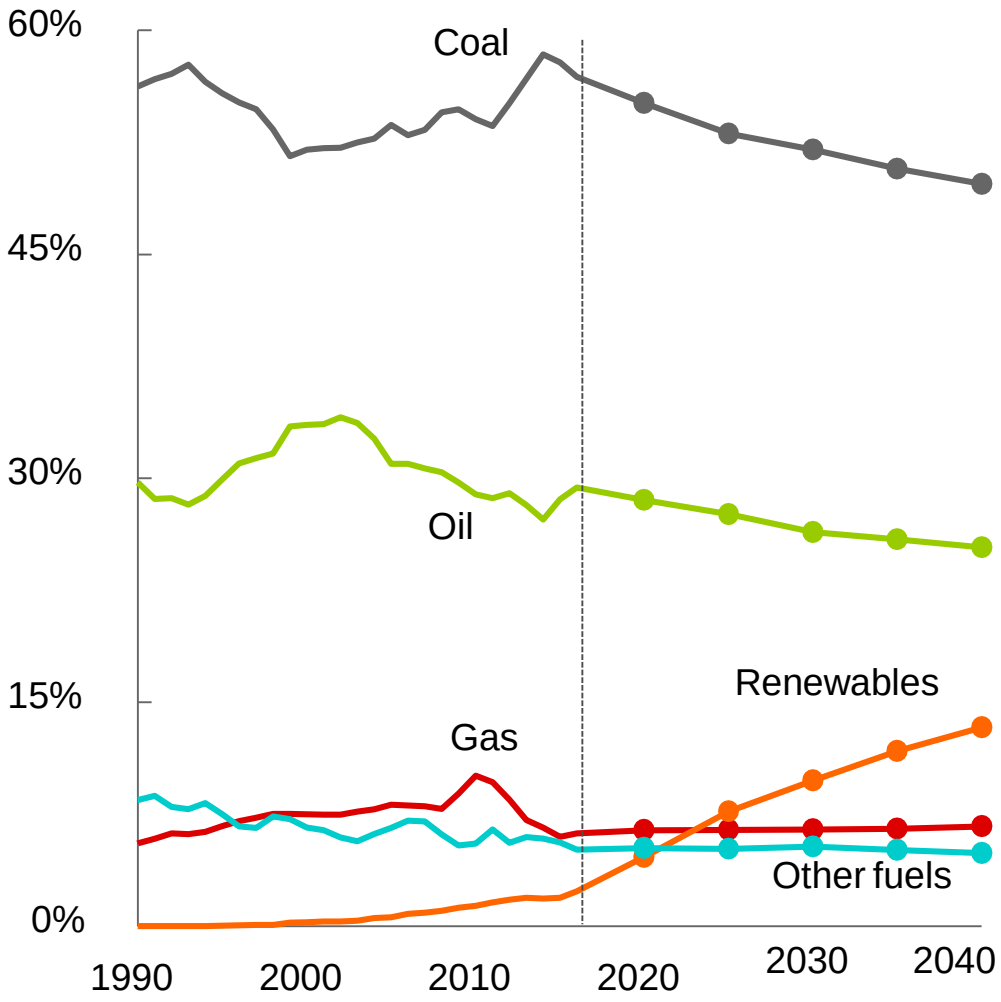
Primary energy demand by fuel and region



BP Energy Outlook-2018

Shares of primary energy

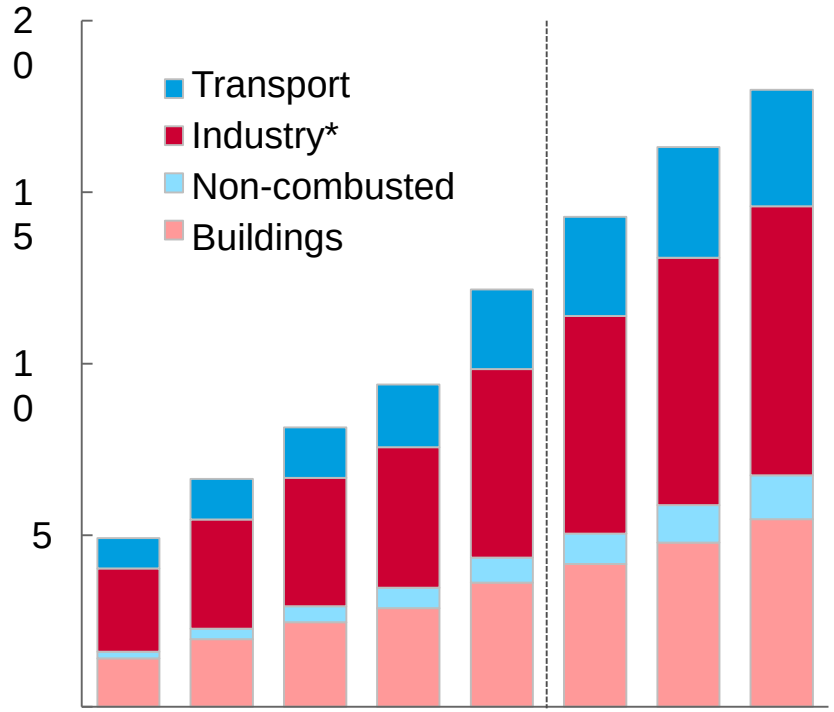
BP Energy Outlook-2018



ENERGY CONSUMPTION BY SECTORS

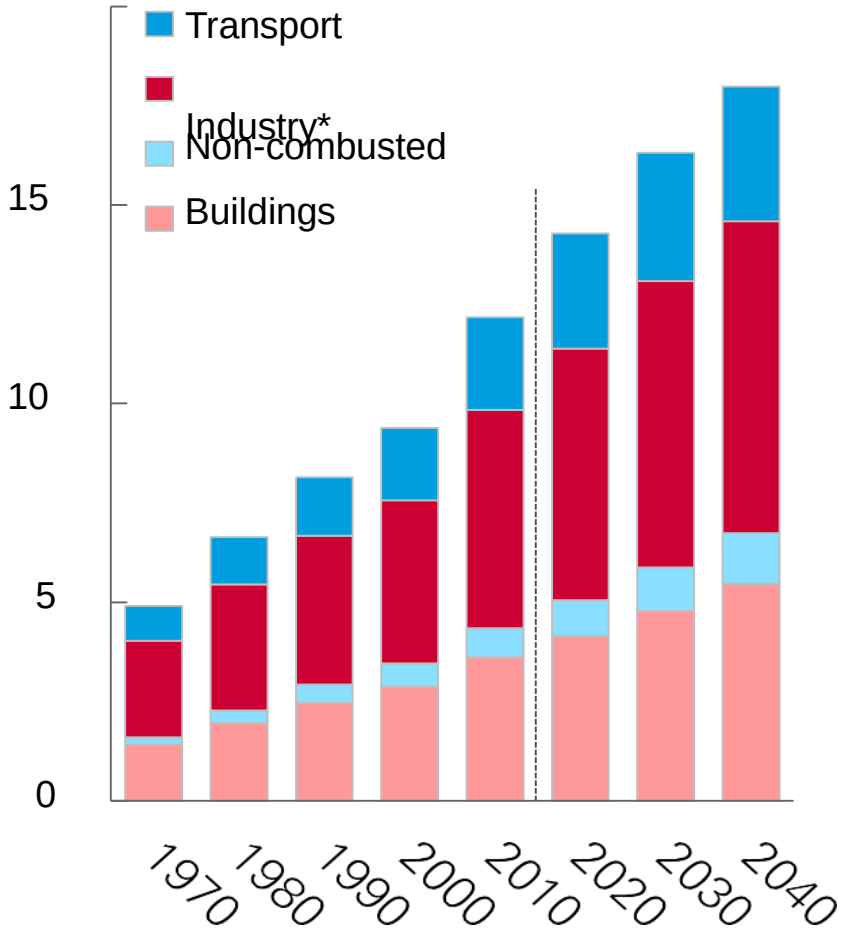
Primary energy consumption by end use sector

Billion toe



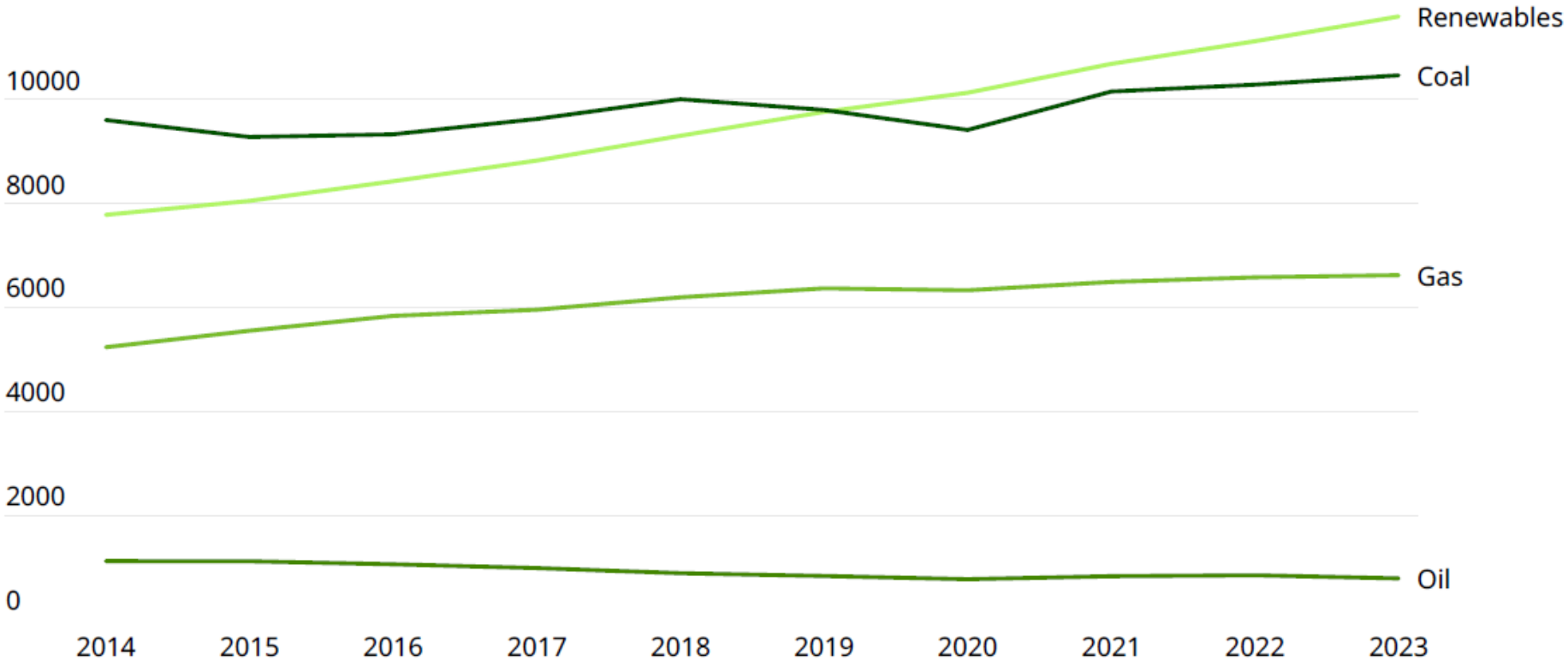
Primary energy demand by end use sector

Billion toe



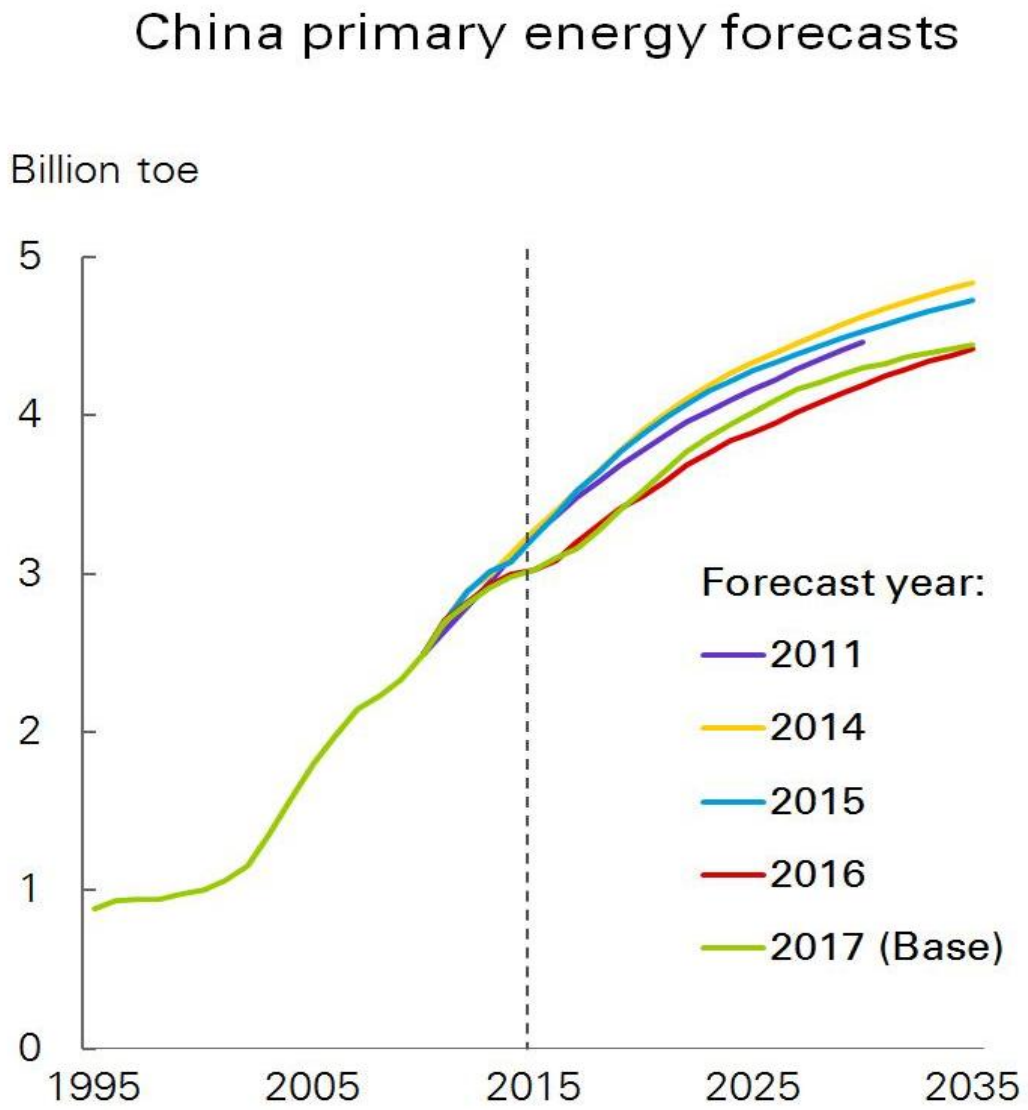
BP Energy Outlook-2018

Global electricity production by source 2014–2025



Source: Ember (2024).

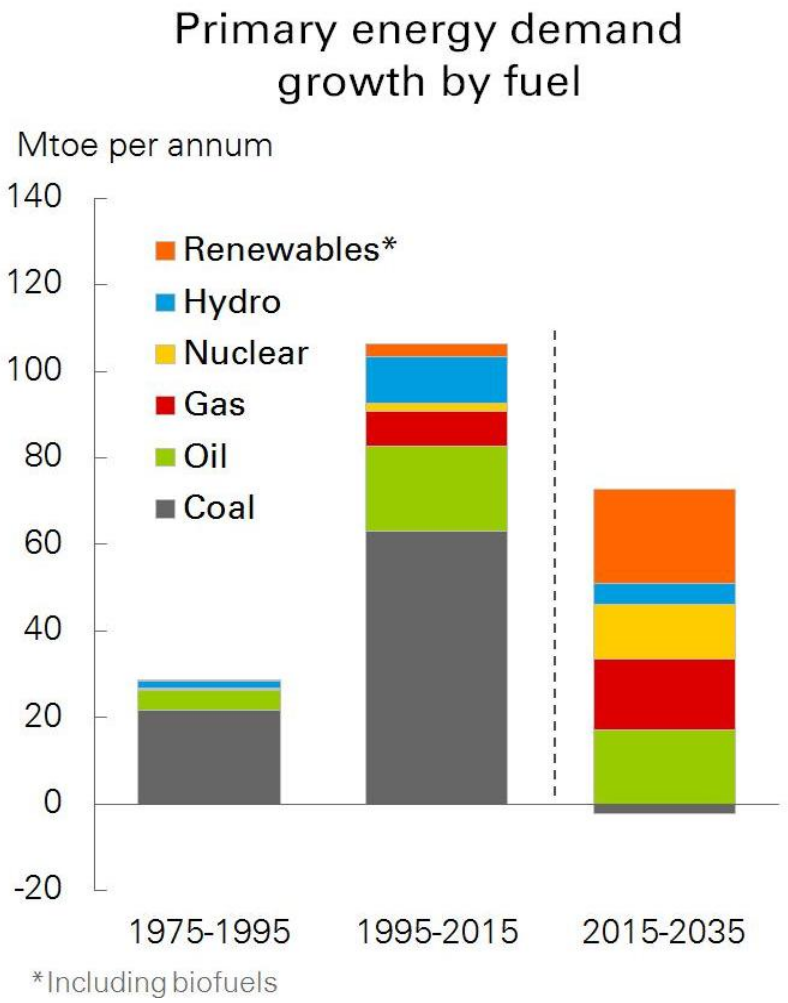
CHINA

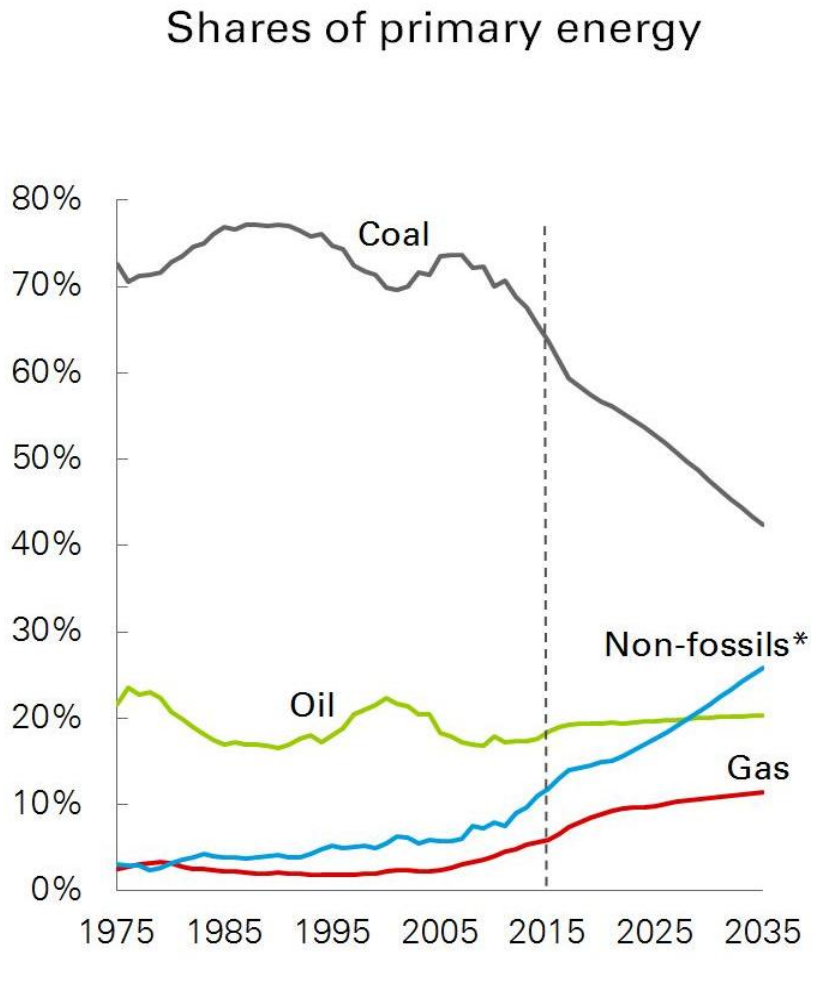


BP Energy Outlook-2017

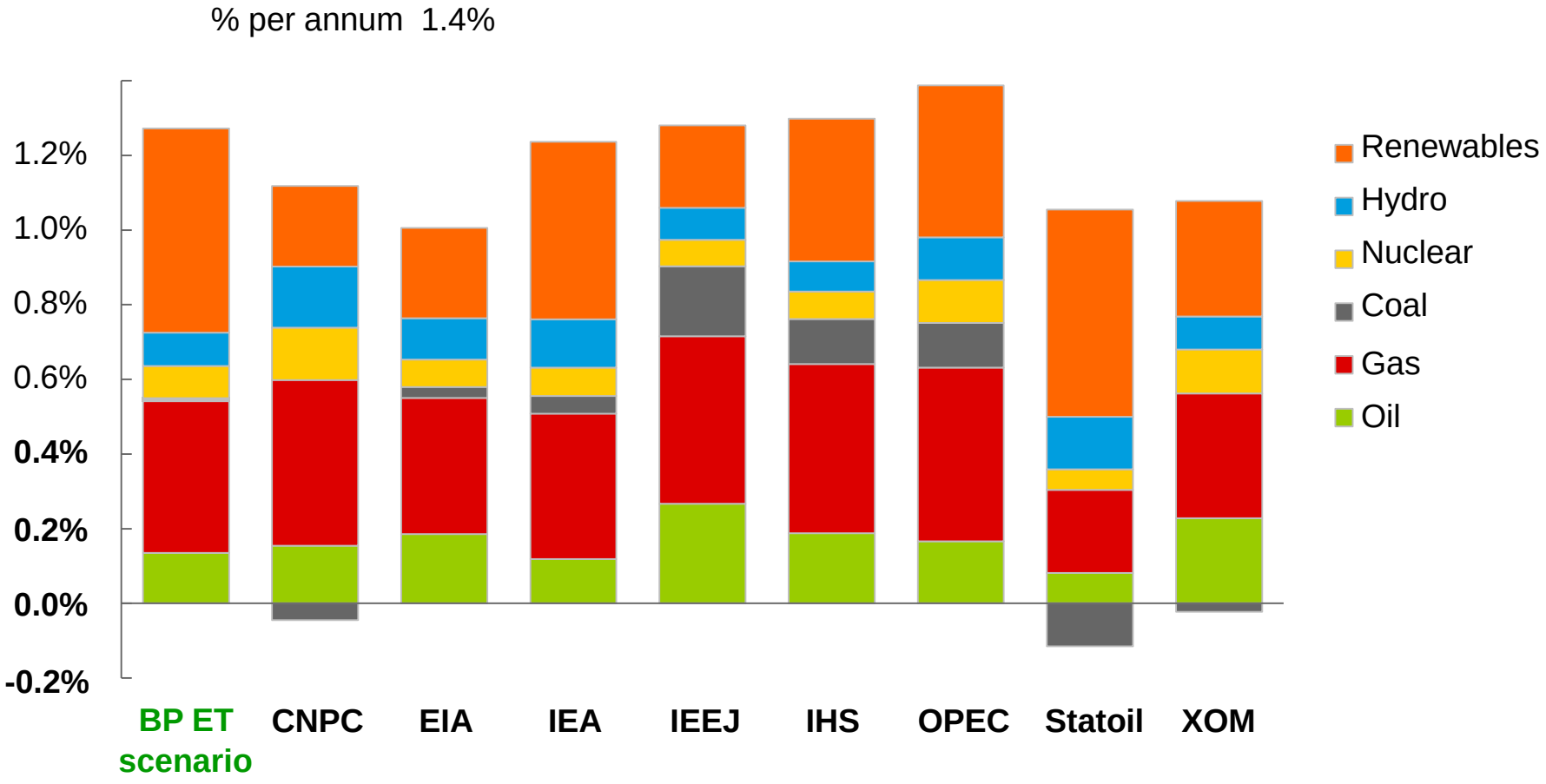
CHINA

BP Energy Outlook-2017

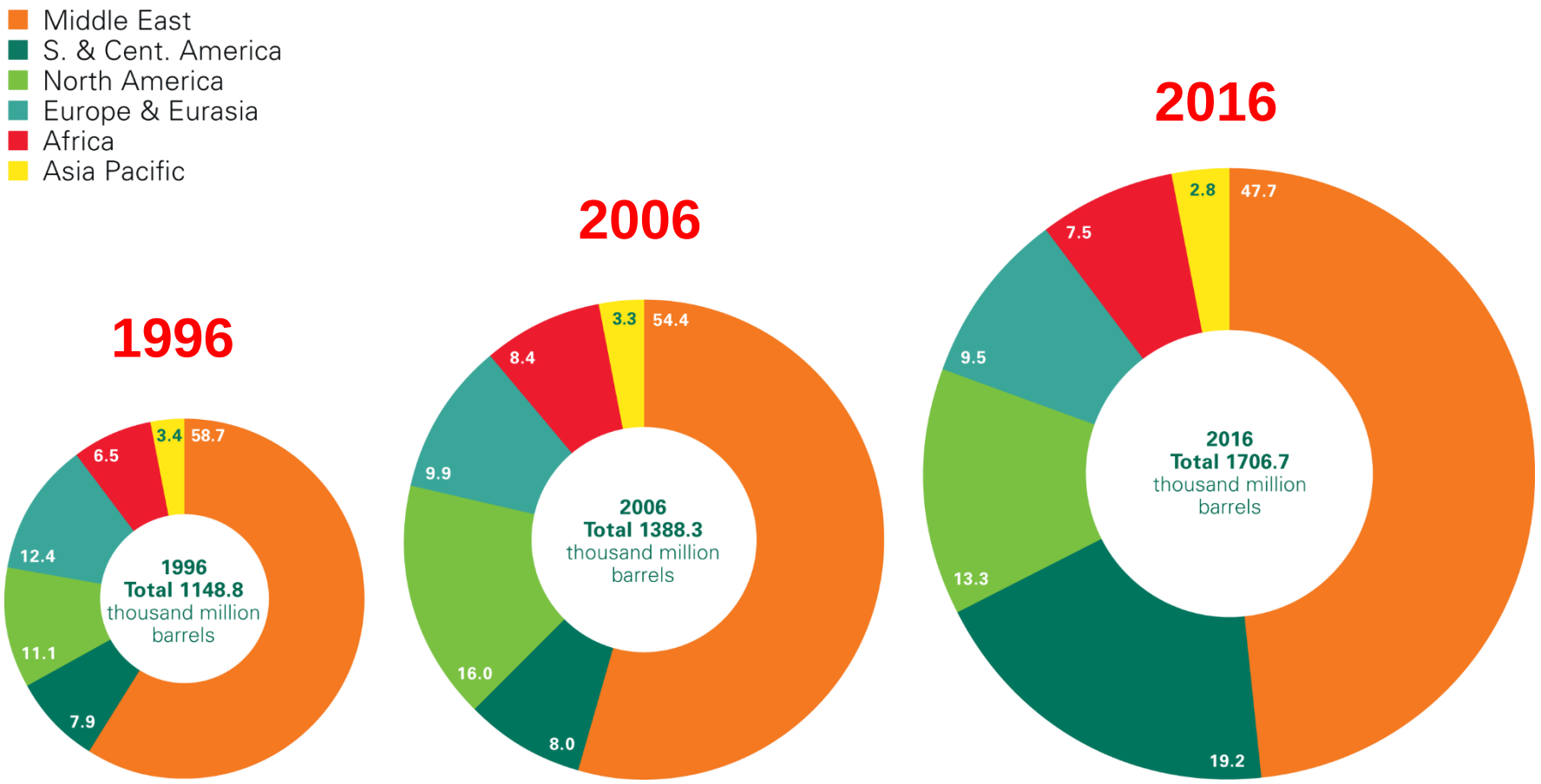




Contributions to growth of energy consumption, 2016-2040



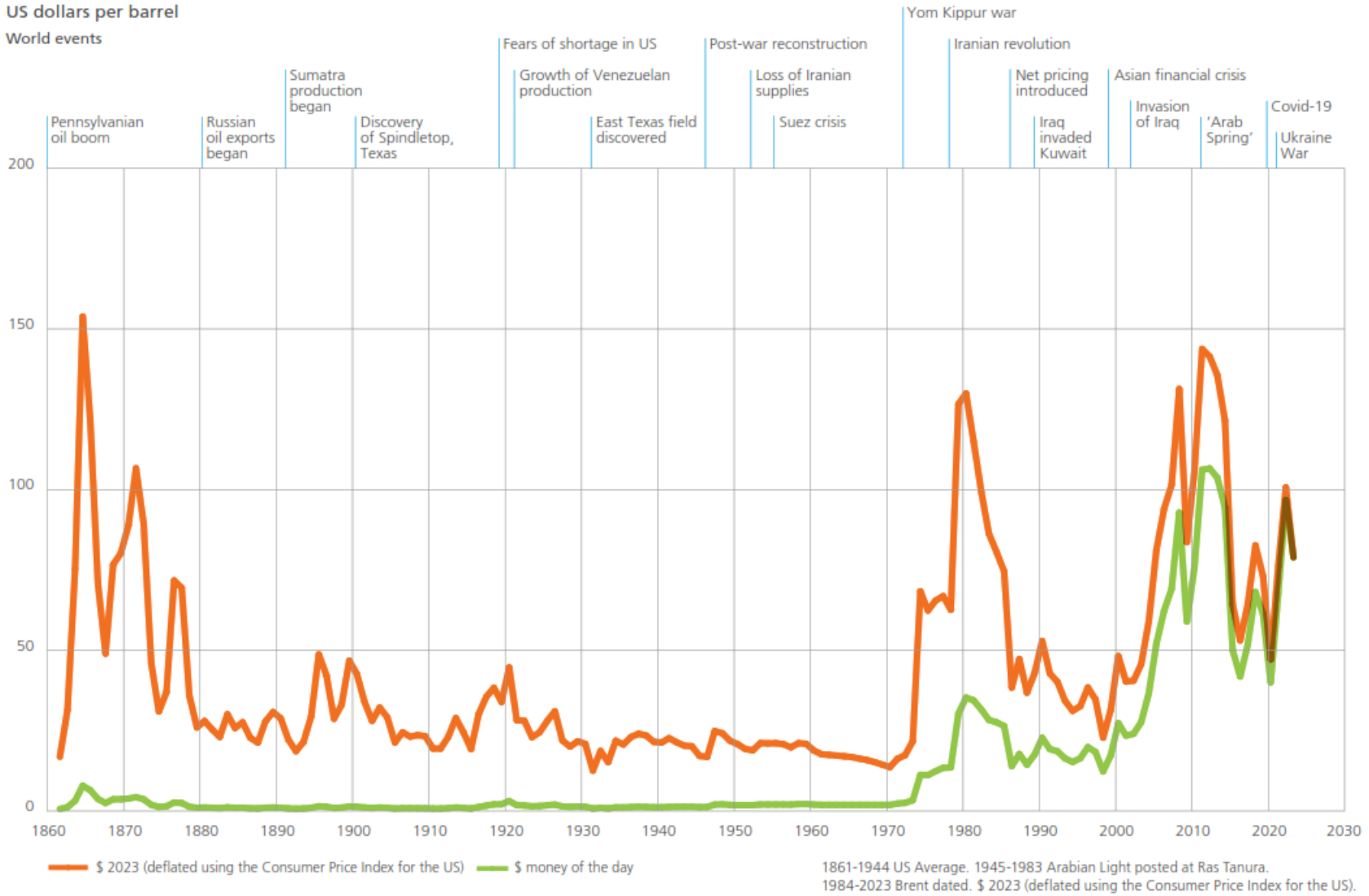
PROVEN OIL RESERVES



BP Energy Outlook-2017

OIL PRICES

Crude oil prices 1861–2023



1861-1944 US Average. 1945-1983 Arabian Light posted at Ras Tanura.
1984-2023 Brent dated. $ 2023 (deflated using the Consumer Price Index for the US).

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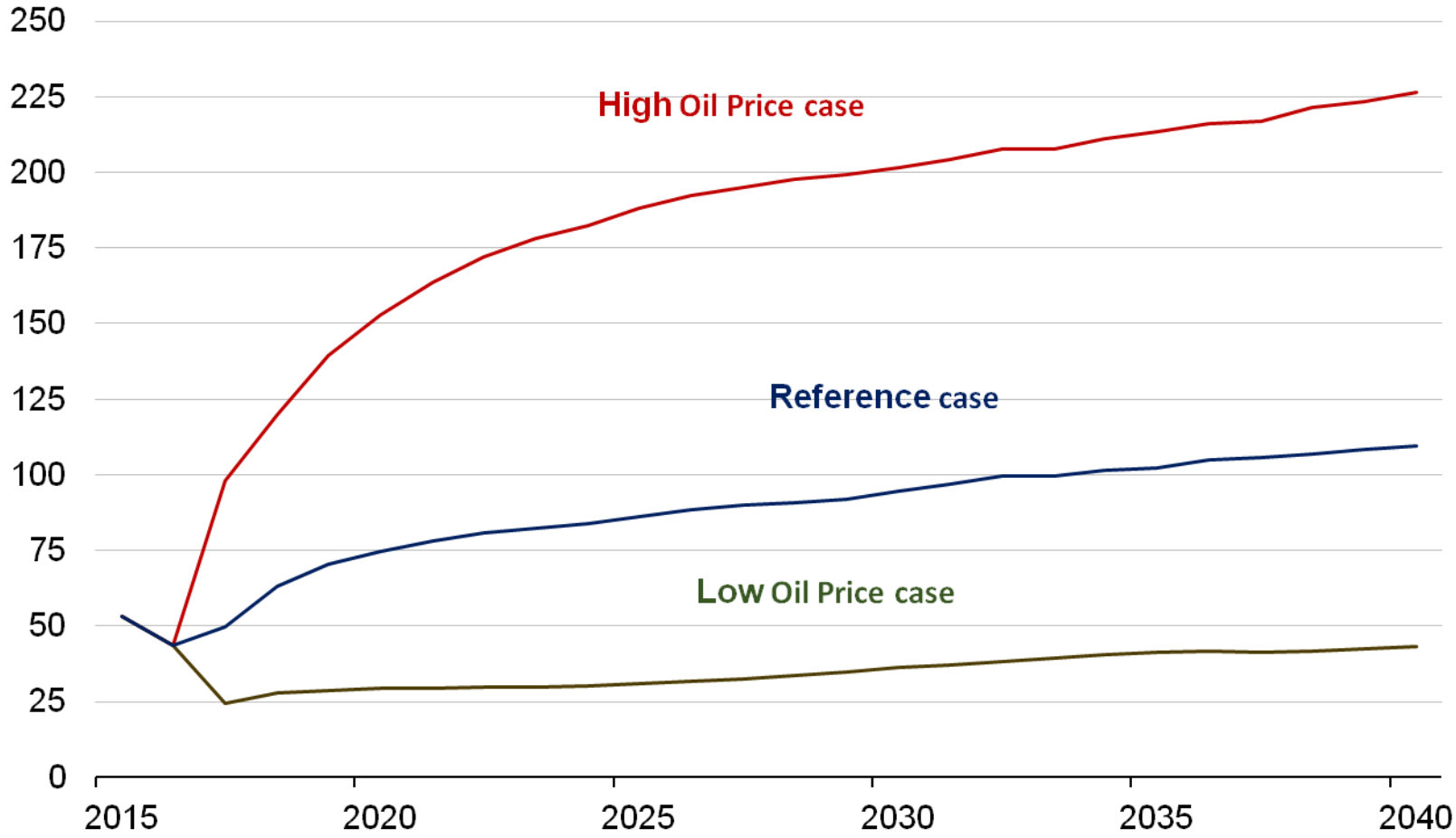
Breakeven crude oil prices

- Breakeven crude oil prices, the price required for producers to cover development and operational costs.

Key Breakeven Price Estimates (approx. 2025–2026)
- **U.S. Shale (New Wells):** $65–$70 per barrel.
- **Offshore Deepwater:** ~$43 per barrel.
- **Onshore Middle East:** ~$27 per barrel.
- **Guyana (Stabroek Block):** ~$30 per barrel.
- **Oil Sands:** $57–$75 per barrel.
- **OPEC+ Fiscal Breakeven:** Saudi Arabia requires ~$96 per barrel to balance its budget, while other nations are lower,
- with the average falling around $72.

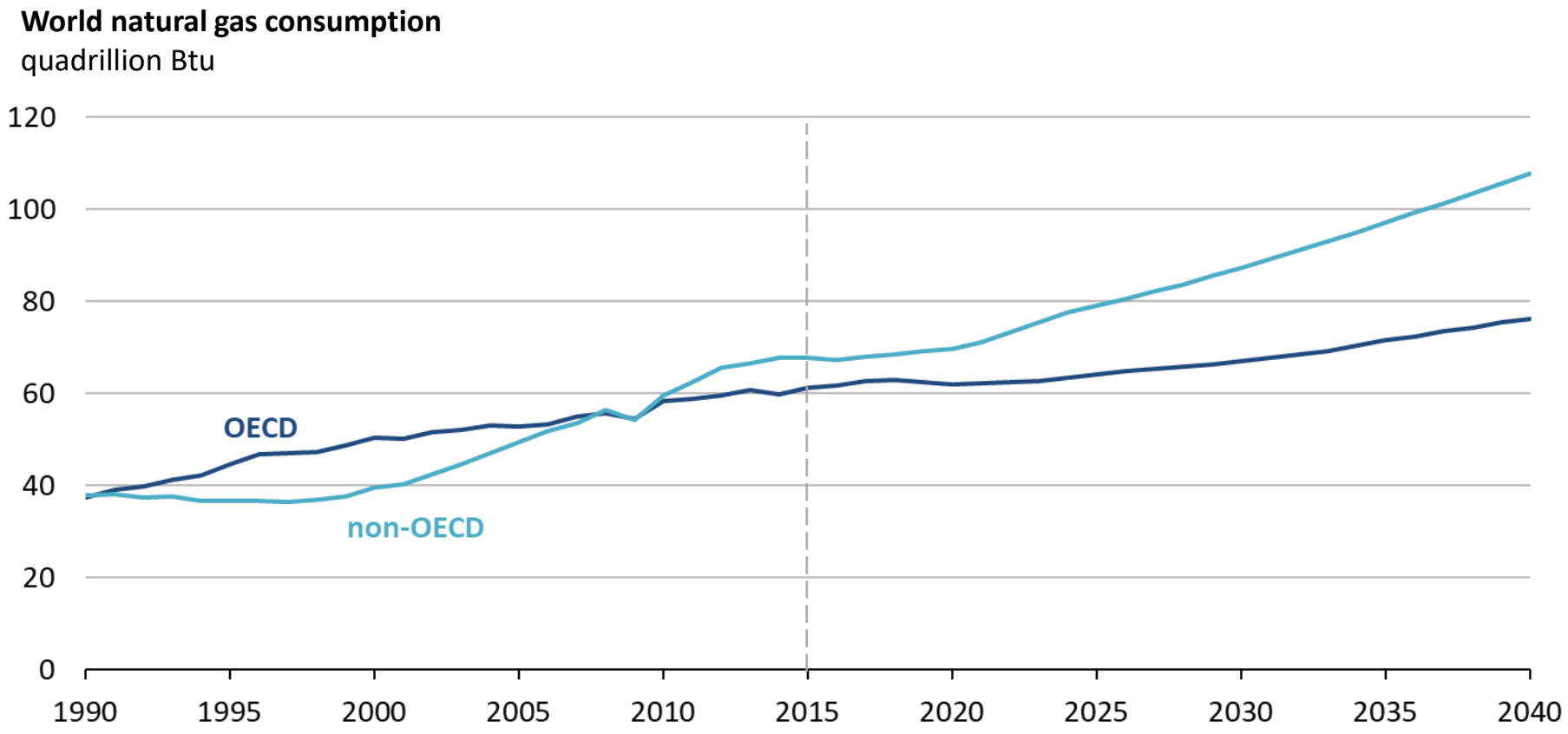
- *Breakeven prices are highly variable based on location, technology, and company efficiency.*

OIL PRICES PROJECTION



IEO-2017, EIA

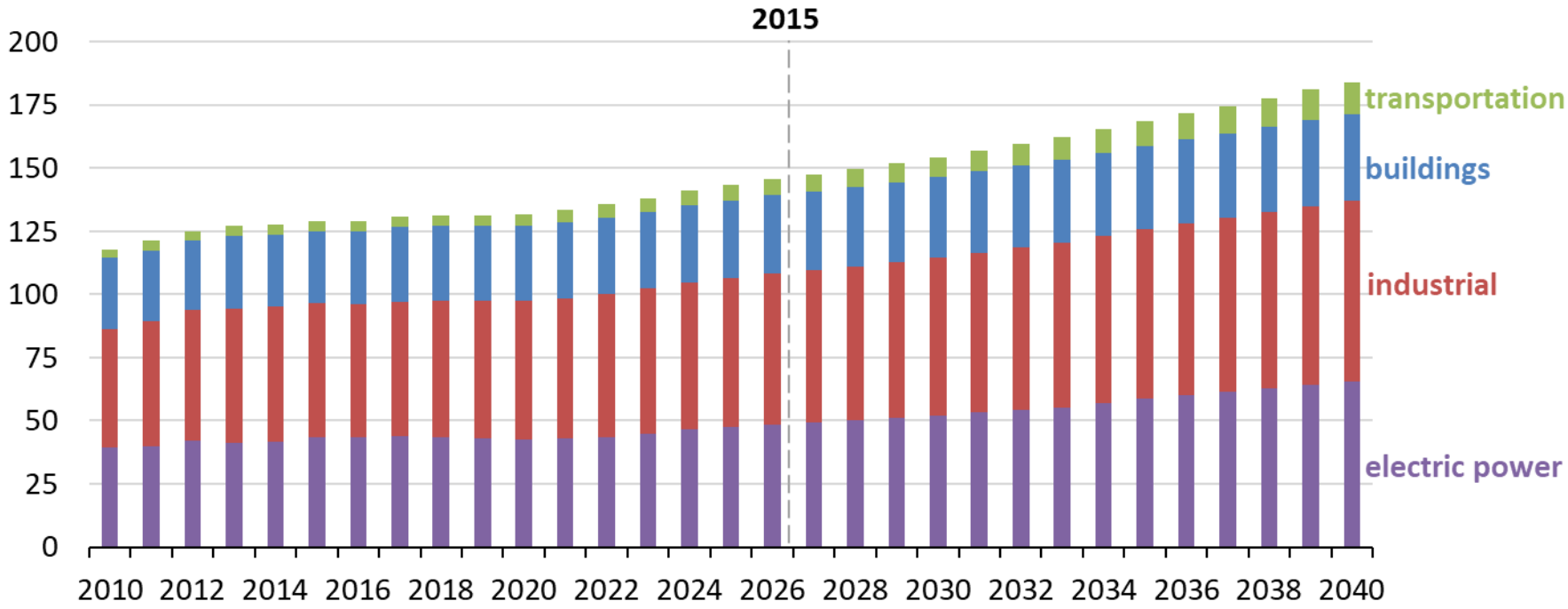
World natural gas consumption



IEO-2017, EIA

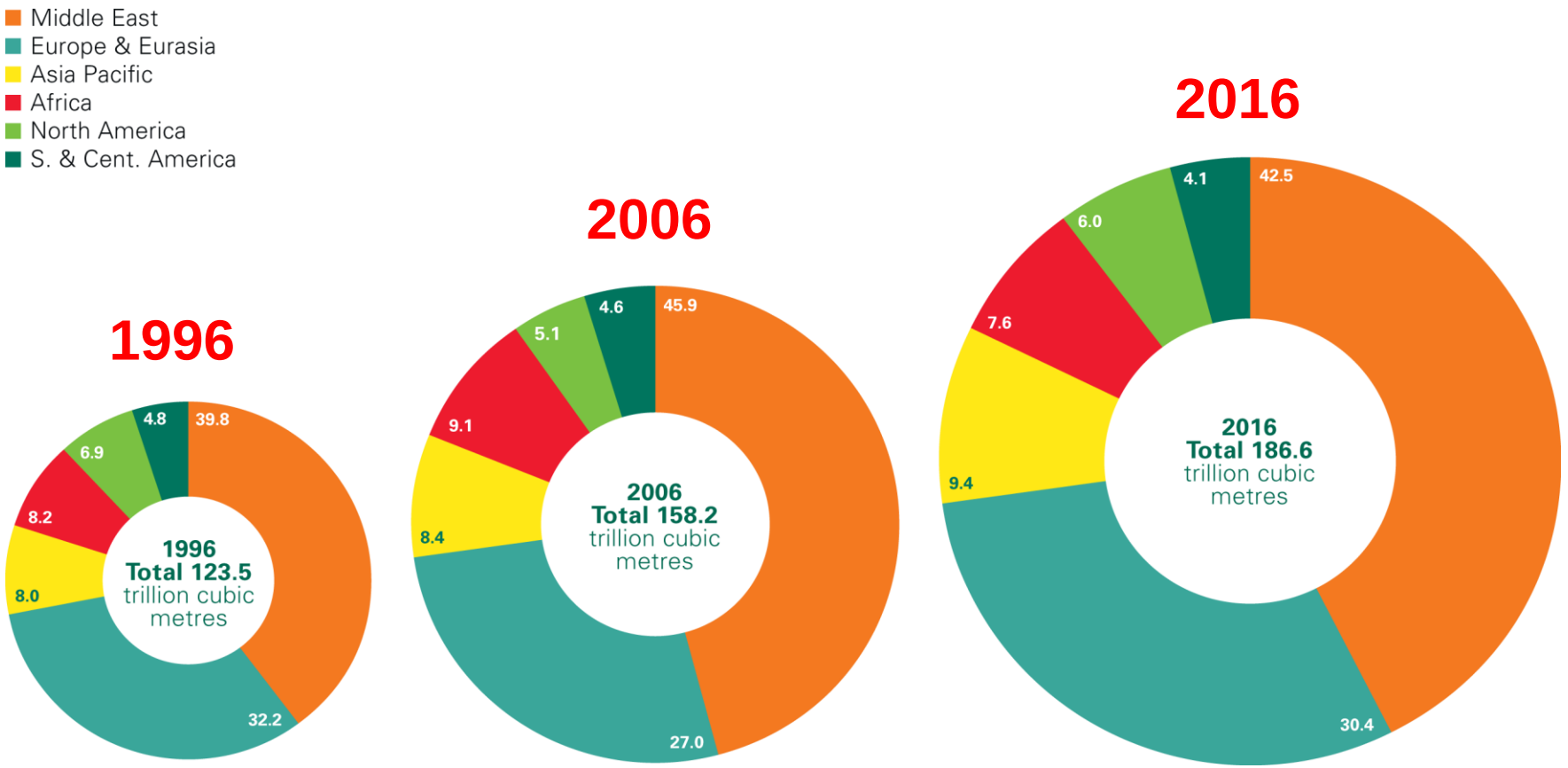
Natural gas consumption increases in all sectors

Natural gas consumption by sector
quadrillion Btu



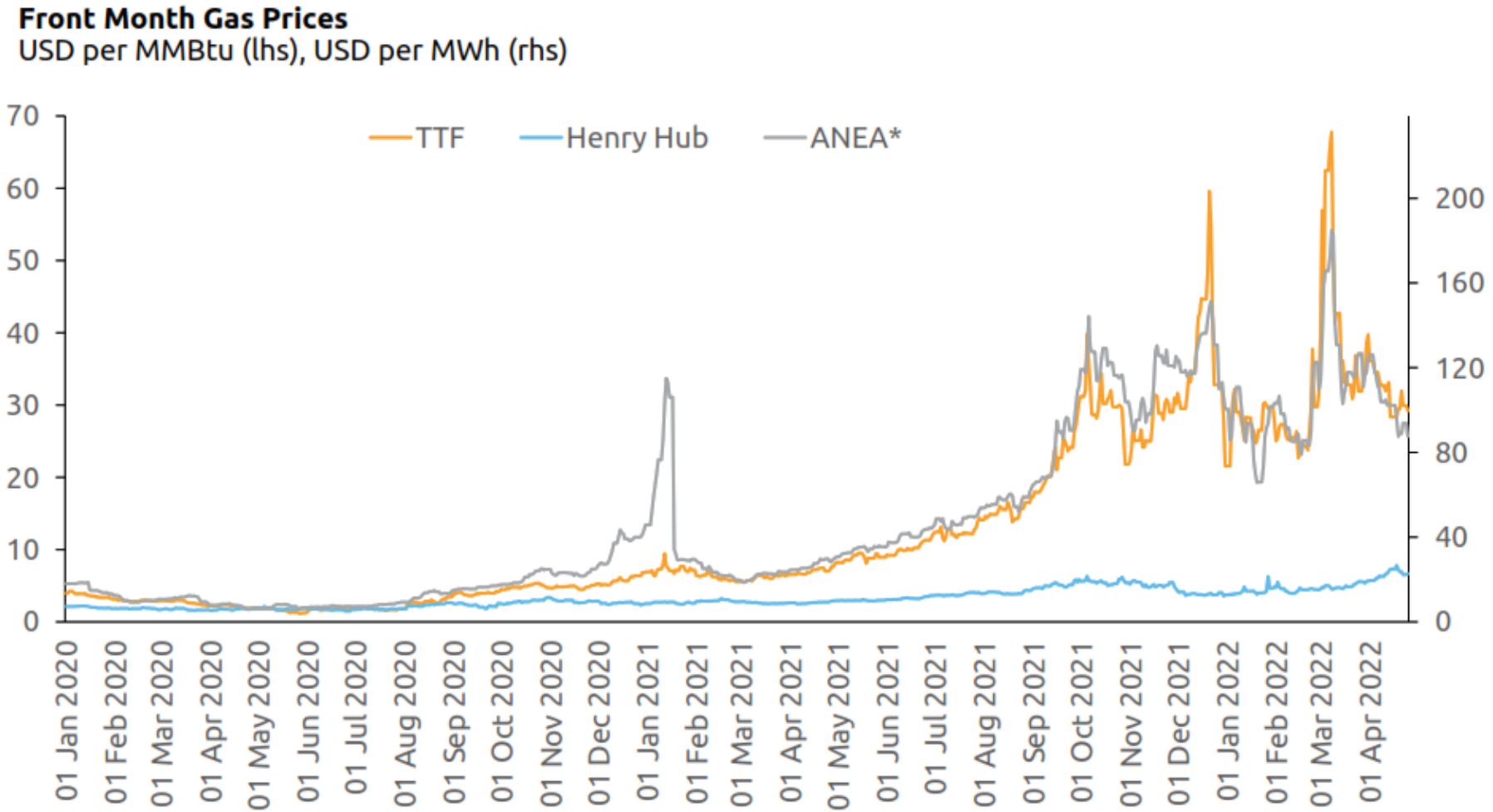
IEO-2017, EIA

PROVEN GAS RESERVES



BP Energy Outlook-2017

Gas prices

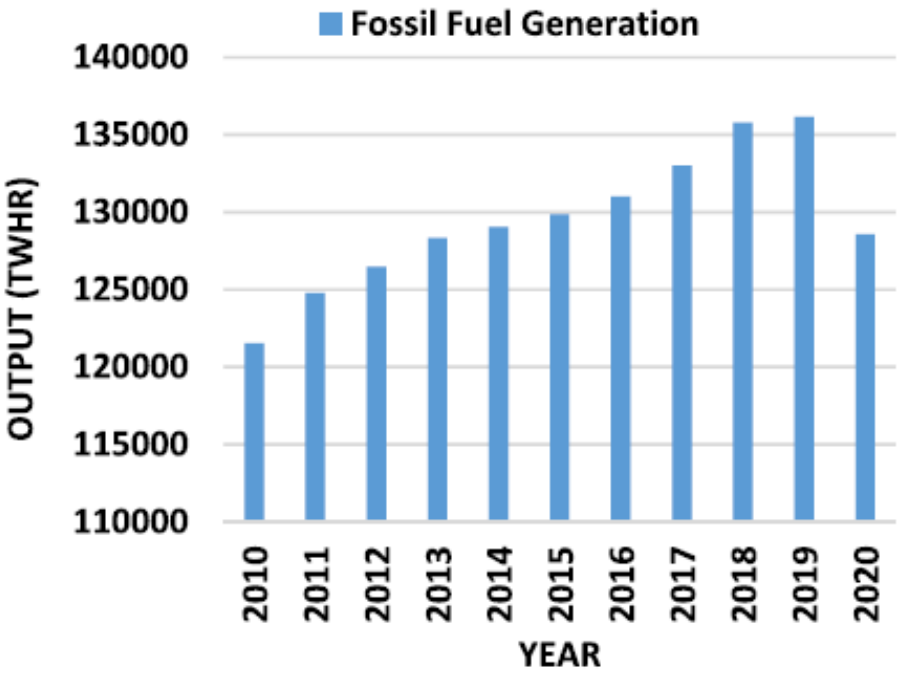


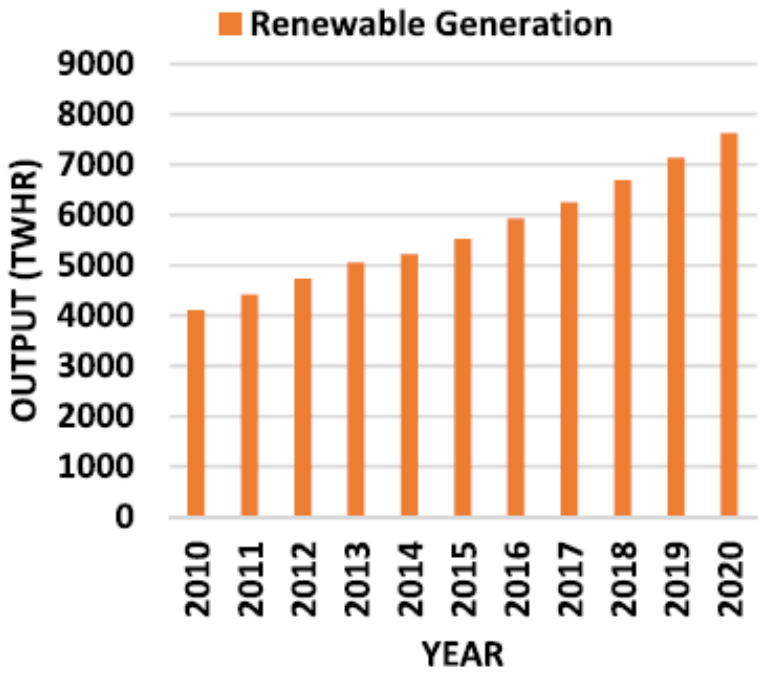
TTF: Title Transfer Facility (Netherlands), ANEA: Argus North East Asia

How long will the energy sources last?

- Oil & Natural Gas – 50 or 75 years? (maybe)
- Coal – 300 years? (maybe)
- Oil shale and tar sands – 350 years? (maybe)
- Nuclear fission
  - Today's lightwater reactors-100 years
  - Future breeders -10,000 years
- Nuclear fusion
  - DT reaction -10,000 years
  - DD reaction ∞
- **Renewables - ∞**

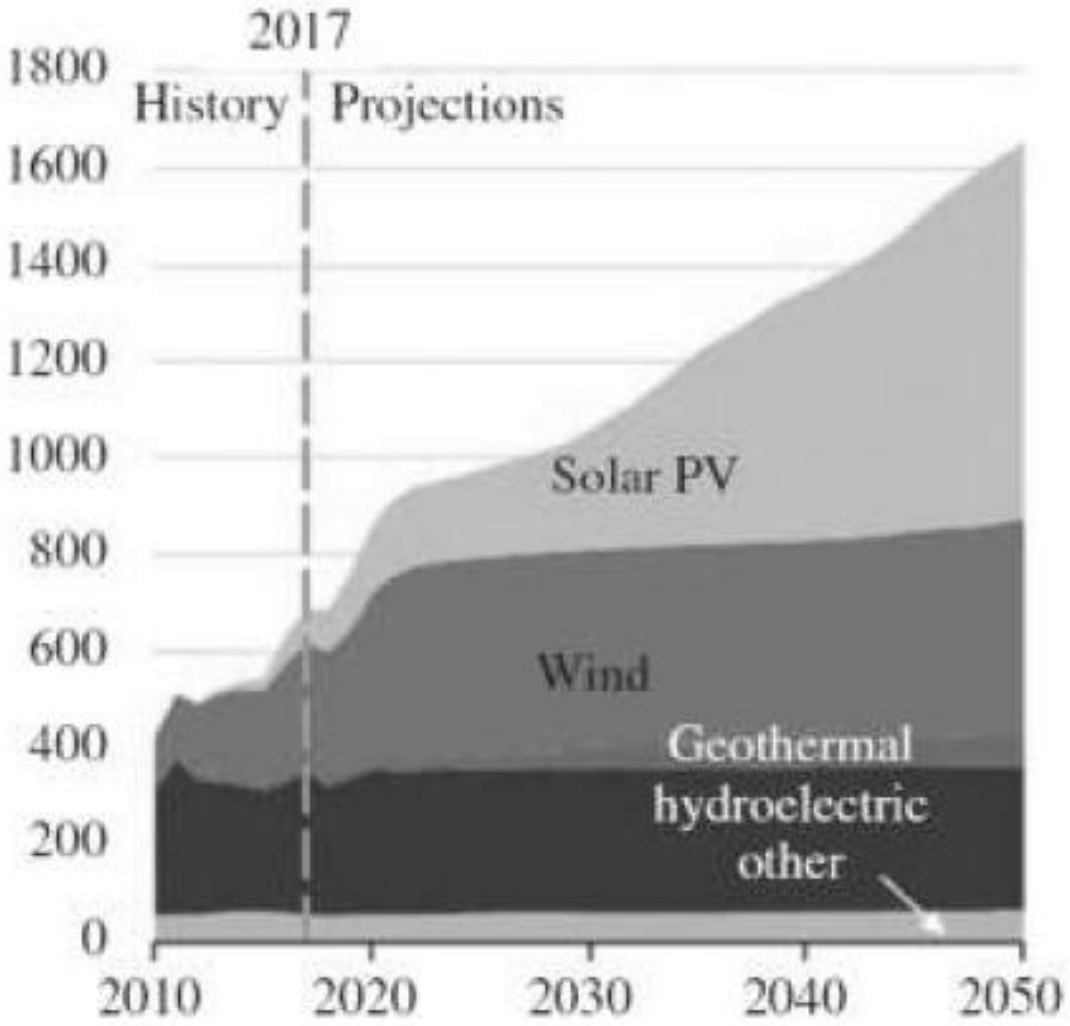
Comparison of fossil fuel generation and renewable generation growth (2010–2020)



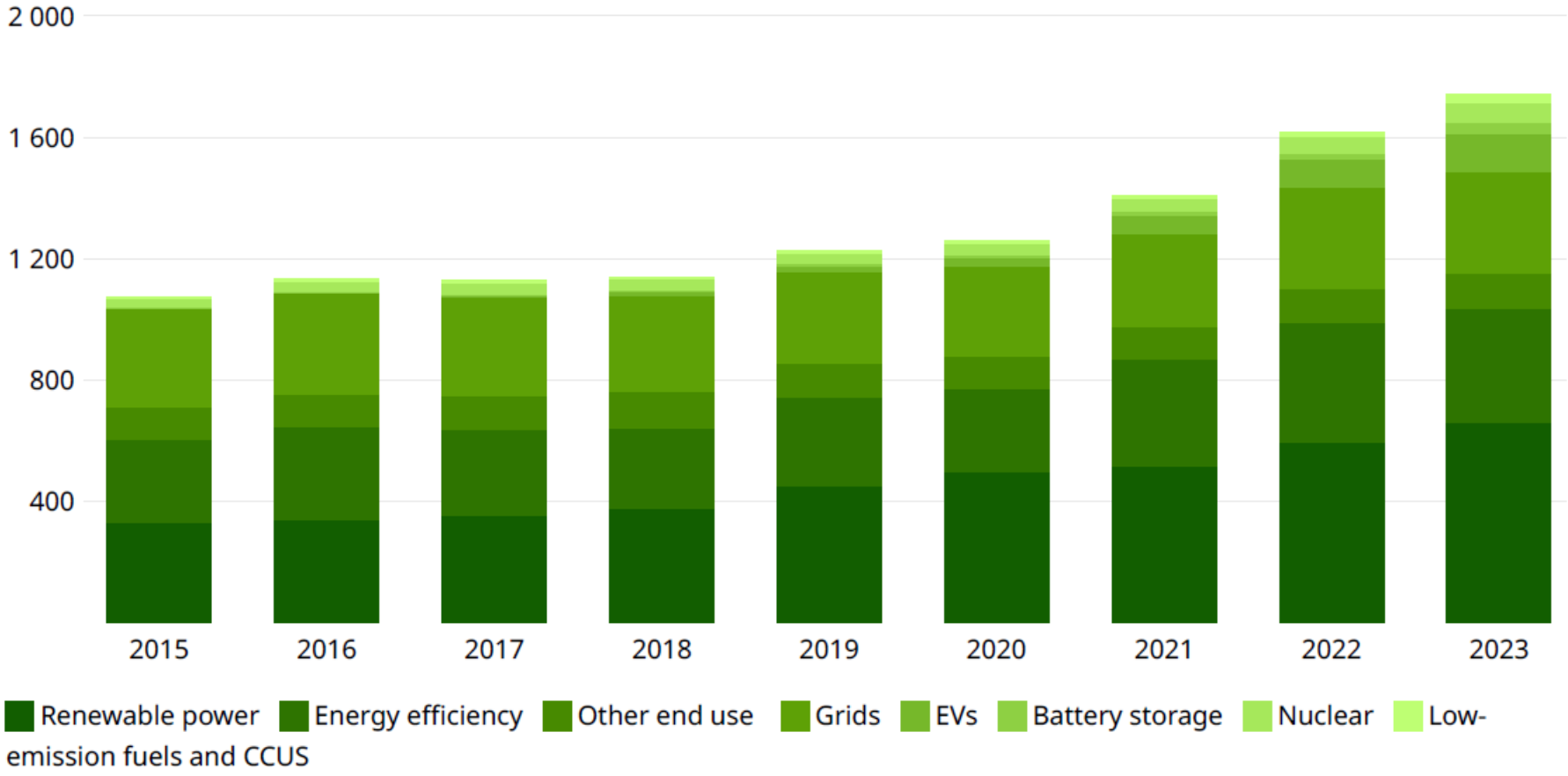


Ang et al. 2022

Renewable electricity generation by source in the United States (billion kWh)



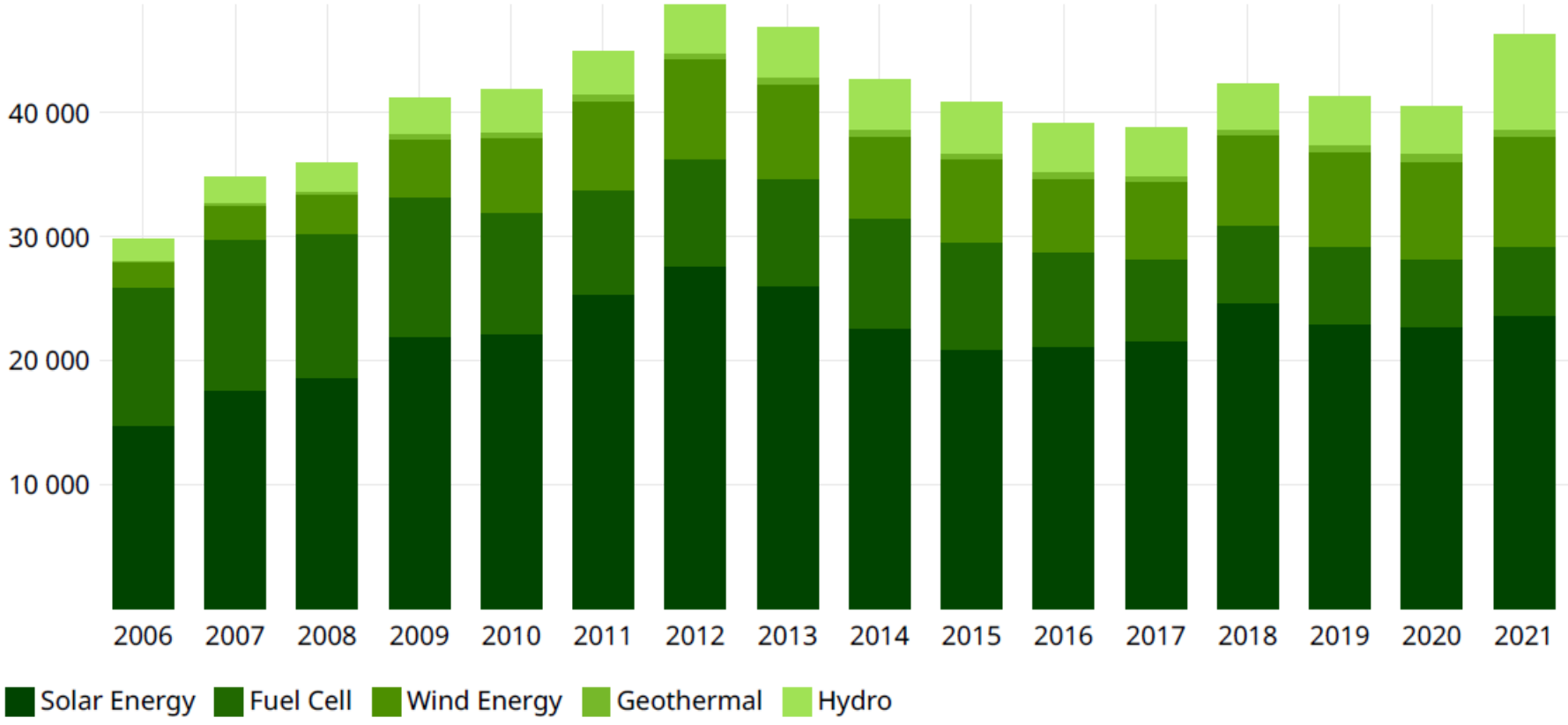
Annual clean energy investments, 2015–2023 (in billion USD)



Note: CCUS = carbon capture, utilization and storage; EVs = electric vehicles.

Source: IEA (2023d).

Trend in patent applications in energy-related technologies, 2006–2021

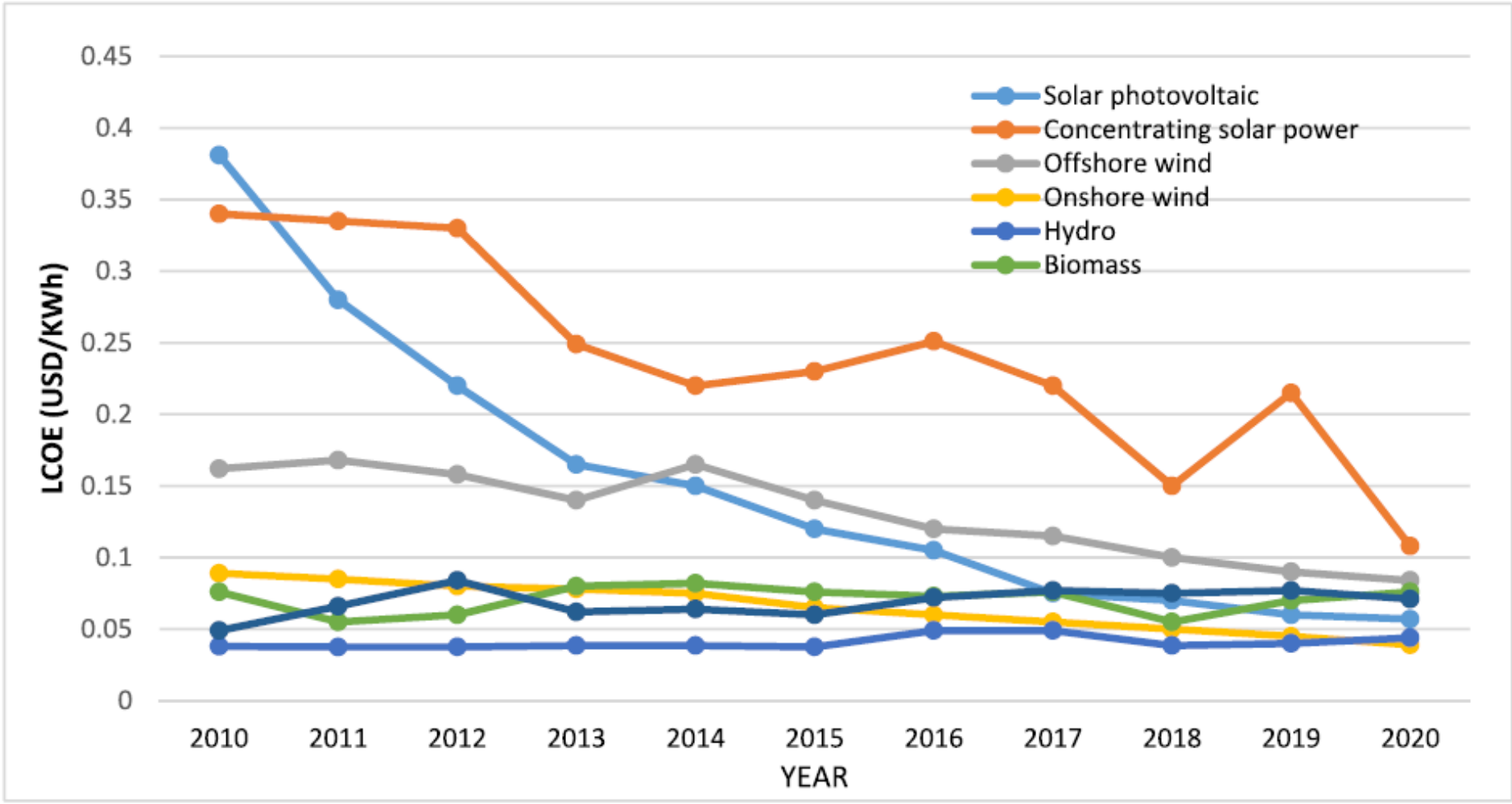


Source: WIPO (2023).

Advantages of Renewable Energy

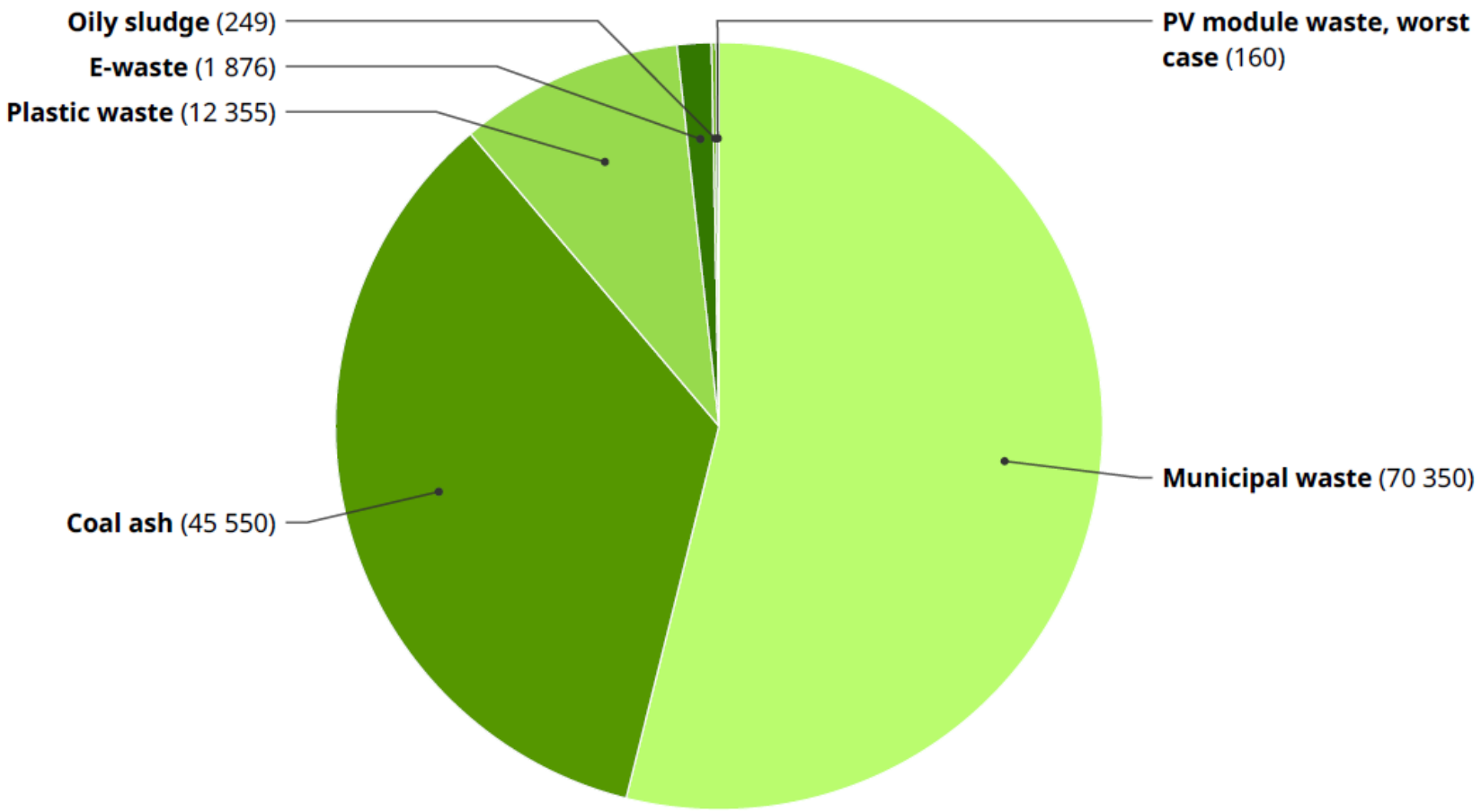
- Natural replenishment,
- Sustainability,
- Being exploited in safe processes,
- Present everywhere,
- Reducing dependence on traditional, non-renewable energy sources
- Less pollution of the environment,
- Reducing the production of CO2 and other greenhouse gases,
- Development of new technologies,
- New jobs in progressive area.

Levelized cost of energy (LCOE)



Ang et al. 2022

Global cumulative waste in 2050 (in million metric tonnes)



Source: Mirletz et al. (2023).


THANK YOU